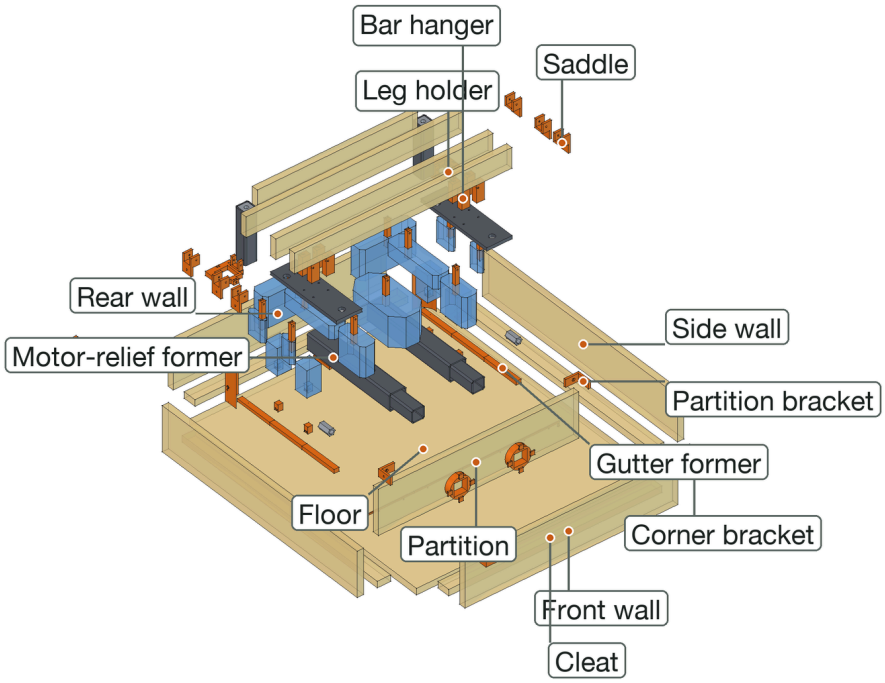
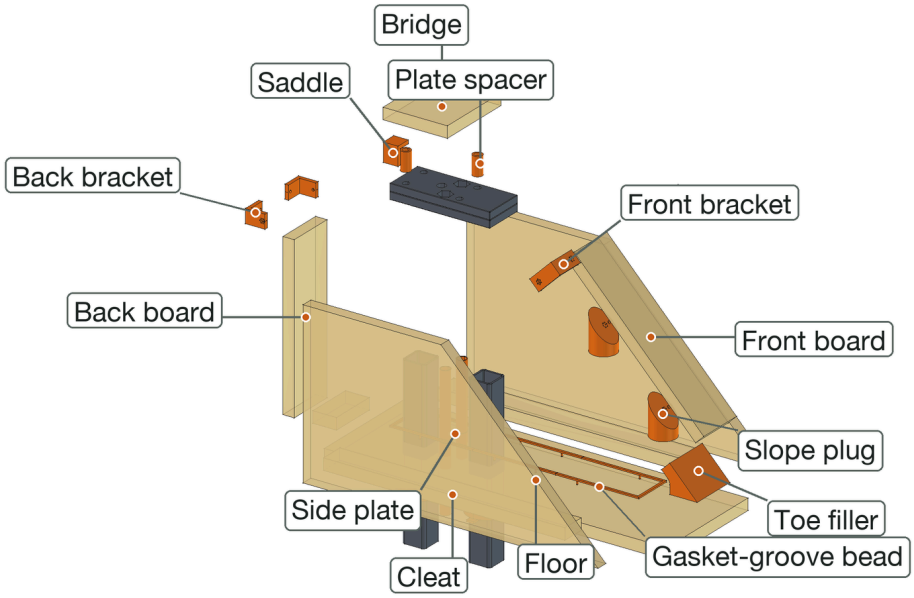
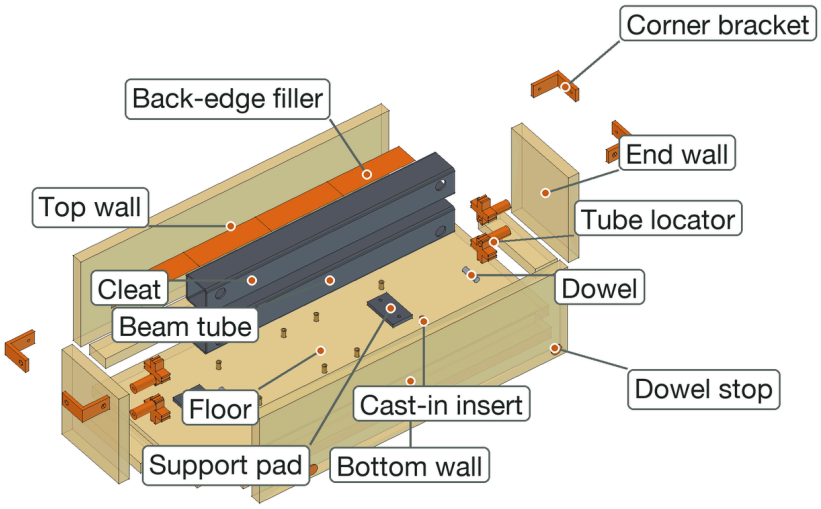
Mold guide

DIY CNC epoxy-granite mill · October 2026

How to make the molds for the five castings. Each mold also has its own shorter guide in this folder (base, uprights, beam), and its own printable drilling templates in `templates/`.





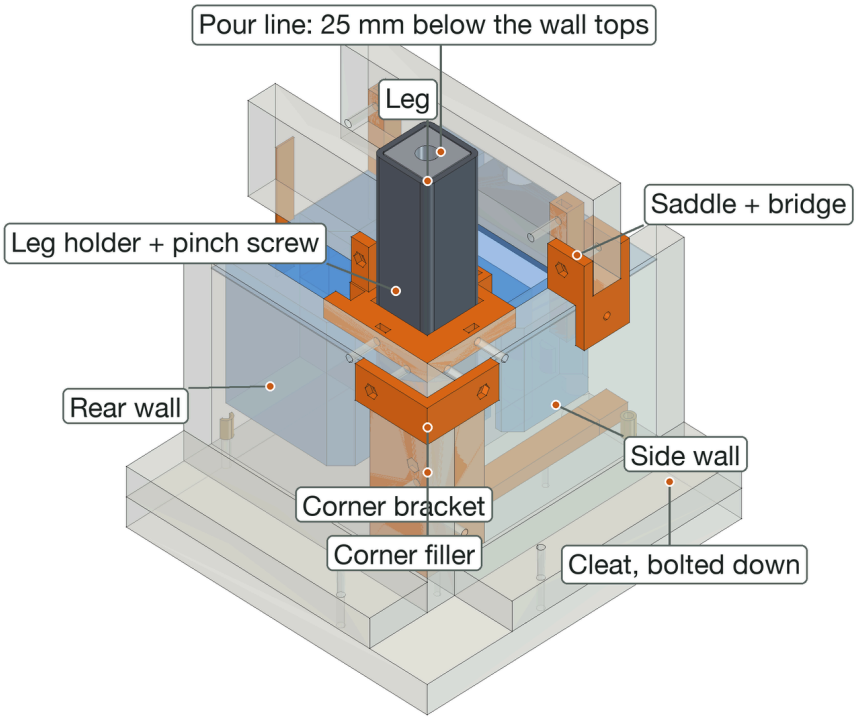


In short: melamine boards bolt together with printed brackets. Every printed piece bolts or pins to its board from outside. The walls stand 25 mm above the pour line, so no bolt goes through a face the mix touches. Nothing is glued, and it all comes apart for the next casting.

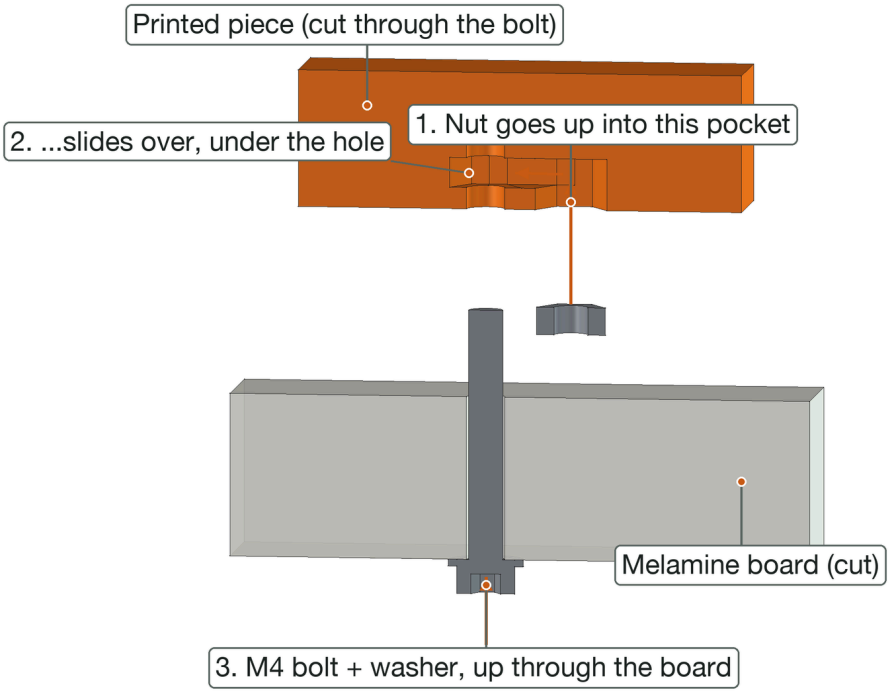
Order of work: print → cut the boards → mark them with the templates and drill → cut the foam → wax → assemble → pour → take apart.

1. How it works

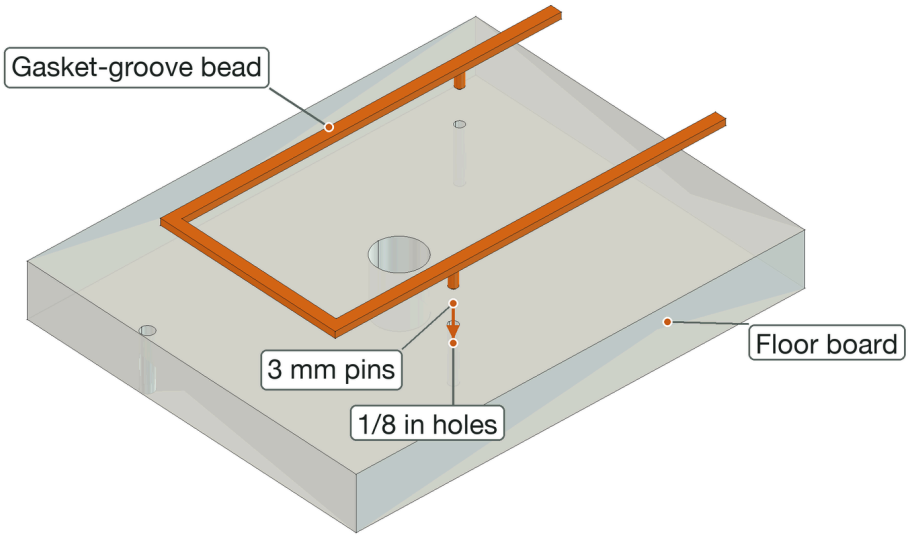
- **Walls** stand on the floor against bolted-down **cleats**; **corner brackets** square the tops. All the bolts go through the top 25 mm, above the pour line.



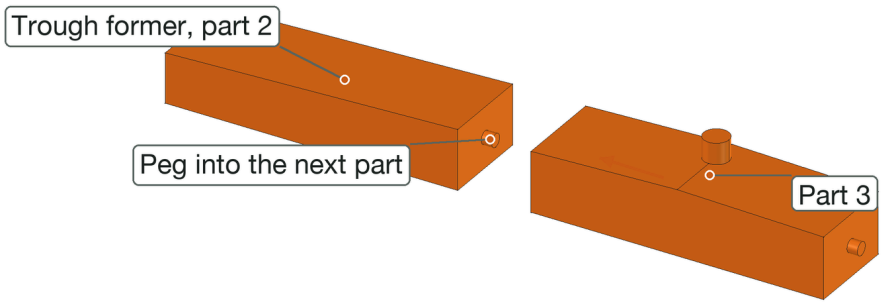
- **Printed pieces** have a nut built in: it goes into a pocket on the board side and slides under the hole; an M4 bolt comes through the board from outside.



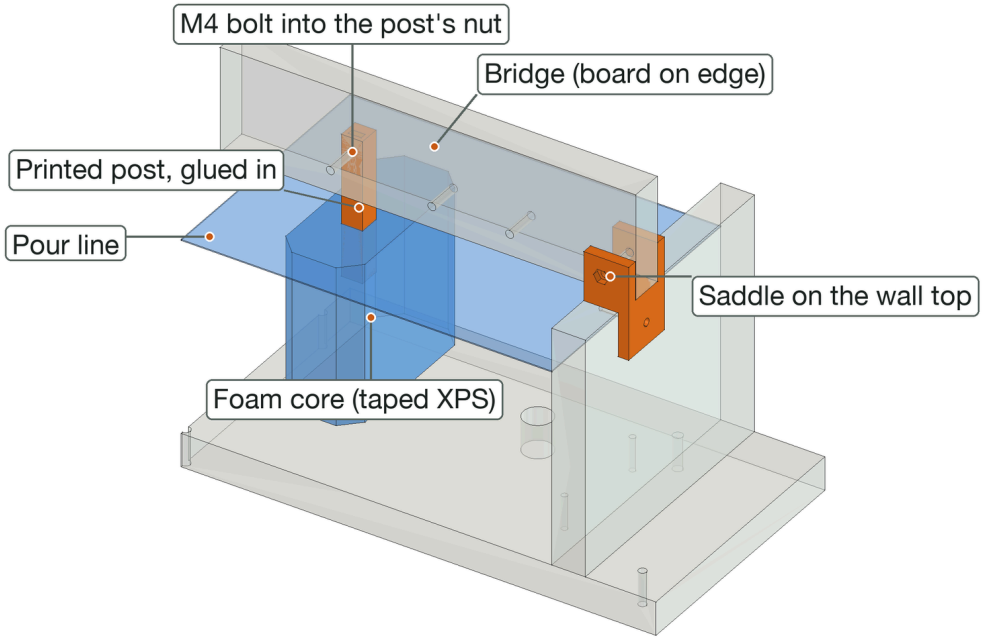
- **Thin pieces** (gasket-groove beads, motor relief) have 3 mm pins that push into 1/8 in holes.



- **Long pieces** are printed in parts that peg together; each part bolts down on its own.



- **Cores** are taped foam blocks with a printed post that bolts to a bridge across the mold.



- **Fits** are 0.15 mm a side, and drawn for 19.05 mm board (Home Depot's 3/4 in melamine is a true 0.750 in). Measure yours; if it differs, set `mold_board` in `model/build_cnc.py` and re-run it.

2. What you need

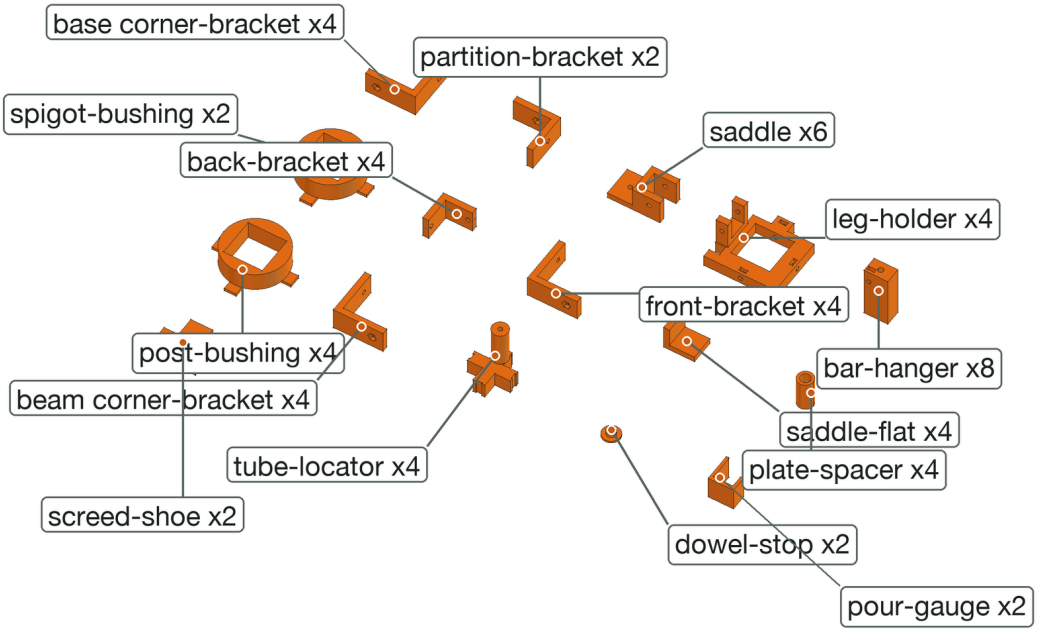
Material	Amount	Notes
Melamine board, 3/4 in	2 sheets, 4 x 8 ft	layout in section 4
XPS foam board, 2 in	1 sheet, 2 x 8 ft	the cores
PETG	1.33 kg	sliced; everything printed for the machine fits in two 1 kg rolls
Packing tape, double-sided carpet tape	1 roll each	wrapping and holding the foam
Paste wax and PVA	1 each	release
3/8 in quarter-round PVC molding	about 6 m	the rounded outside edges
2x4 offcut	4 blocks per upright mold, about 50 mm	feet

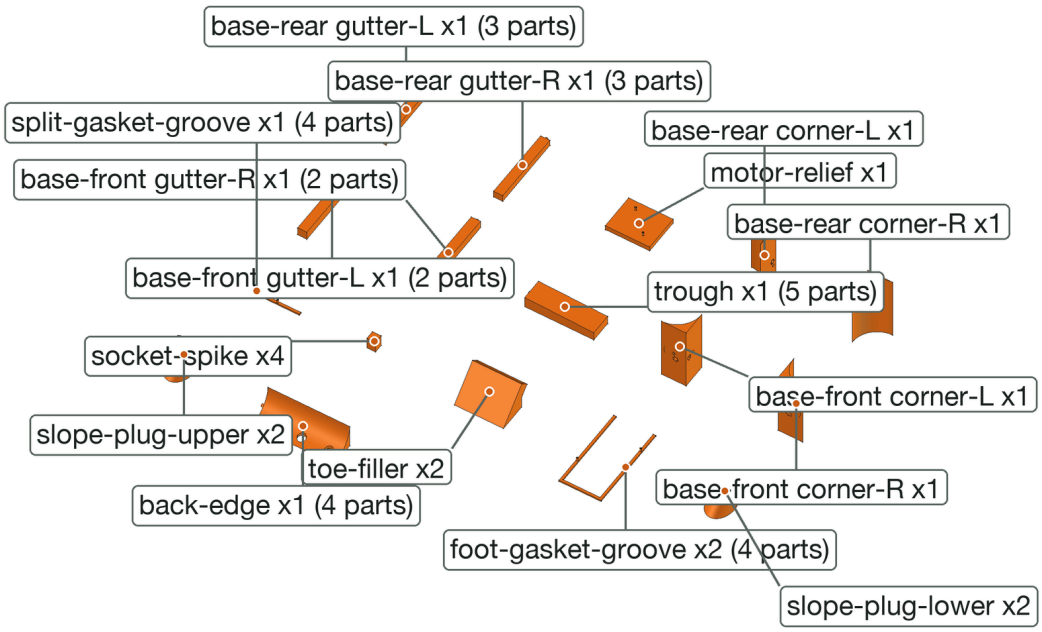
Hardware	Qty
M3 nut	20
M3 washer	20
M3 x 25 bolt	20
M4 nut	190
M4 washer	186
M4 x 20 bolt (leg pinch screw)	4
M4 x 30 bolt	114
M4 x 35 bolt	28
M4 x 45 bolt	40
M4 x 70 bolt	4
M5 x 20 screw, greased (holds an insert)	29
M5 x 60 screw (hangs an anchor bar)	8
M6 x 20 screw, greased (holds a pad)	6
M6 x 20 screw, greased (holds an insert)	6
1/2 in plain steel rod, greased (forms a beam bolt hole), 198 mm	8
1/2 in rod (or a tie rod), greased, 381 mm: centers a top tie-rod sleeve	4
1/2-13 x 1-1/2 in bolt + nut, greased (forms the dowel hole)	2
1/2-13 x 1-1/2 in bolt, greased (holds a draw-rod coupler)	2
3/8-16 x 3 in bolt, greased (holds the top plate, both layers)	4
5/8 in plain steel rod, greased (forms a slope tie-rod hole), cut to 134 mm, 237 mm	2
5/8 in plain steel rod, greased (forms a tie-rod hole through the base and the anchor bar), 169 mm	6

Add about 10 % spare small bolts, nuts and washers. **Tools:** drill and brad-point bits (sizes in section 5), a 2-1/2 in hole saw, a jigsaw for the side plates' slope, a square, clamps, a 7 mm nut driver, a 3 mm hex key, an awl, a fine saw or hot wire for the foam.

3. Print

150 prints, 1.33 kg, about 115 hours (sliced; settings in the print guide). Print one corner bracket and one saddle first and check them on a board offcut. Press or slide the nuts in before the pieces go on the boards.





File	Qty	g	Time each
`base-box/kit-corner-bracket.stl`	4	33	39m 23s
`base-box/kit-partition-bracket.stl`	2	14	35m 28s
`base-box/kit-saddle.stl`	6	58	46m 3s
`base-box/kit-leg-holder.stl`	4	82	1h 37m 1s
`base-box/kit-bar-hanger.stl`	8	84	56m 45s
`base-rear/former-gutter-L-part1of3.stl`	1	8	35m 5s
`base-rear/former-gutter-L-part2of3.stl`	1	8	33m 40s
`base-rear/former-gutter-L-part3of3.stl`	1	7	31m 15s
`base-rear/former-gutter-R-part1of3.stl`	1	8	35m 6s
`base-rear/former-gutter-R-part2of3.stl`	1	8	33m 40s
`base-rear/former-gutter-R-part3of3.stl`	1	7	31m 15s
`base-rear/former-motor-relief.stl`	1	12	40m 58s
`base-rear/filler-corner-L.stl`	1	17	1h 44m 24s
`base-rear/filler-corner-R.stl`	1	16	1h 42m 27s
`base-front/former-gutter-L-part1of2.stl`	1	11	49m 23s
`base-front/former-gutter-L-part2of2.stl`	1	10	43m 43s
`base-front/former-gutter-R-part1of2.stl`	1	11	49m 24s
`base-front/former-gutter-R-part2of2.stl`	1	10	43m 43s
`base-front/former-trough-part1of5.stl`	1	20	1h 14m 18s

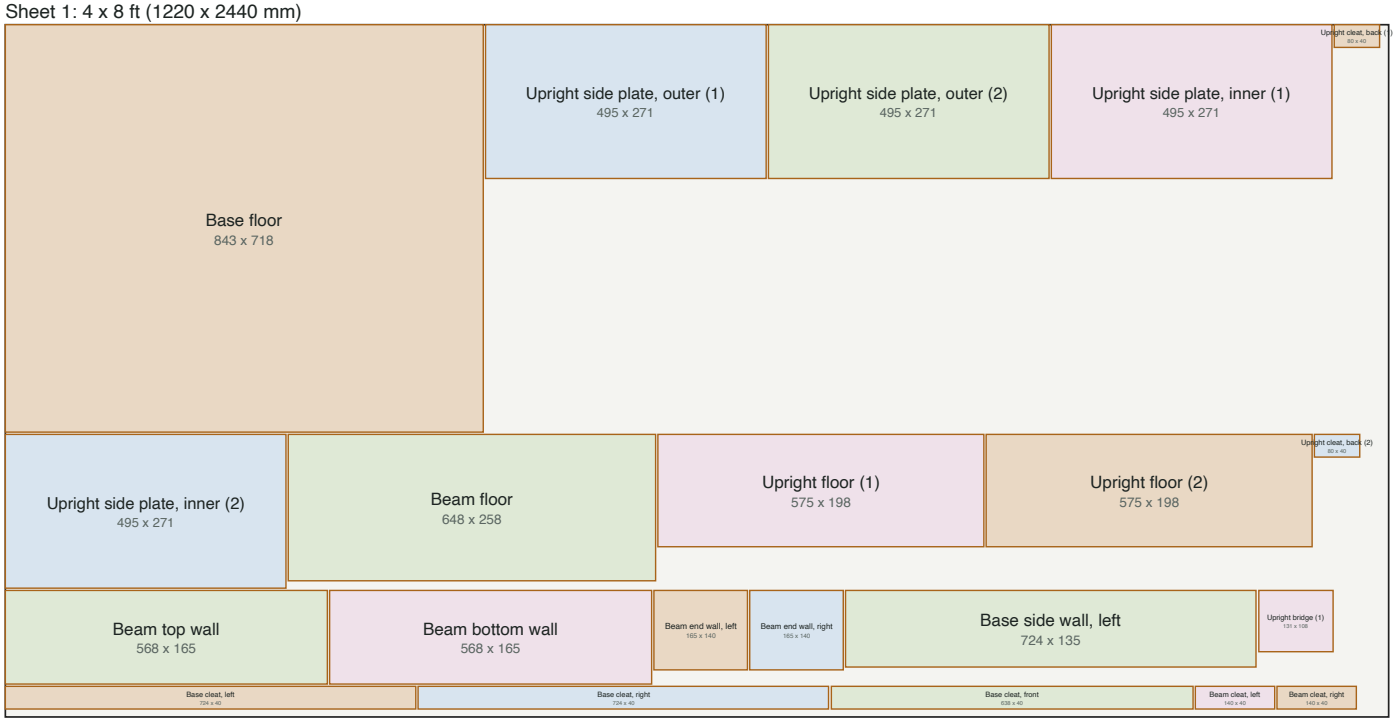
File	Qty	g	Time each
base-front/former-trough-part2of5.stl	1	22	1h 23m 56s
base-front/former-trough-part3of5.stl	1	25	1h 33m 33s
base-front/former-trough-part4of5.stl	1	23	1h 24m 3s
base-front/former-trough-part5of5.stl	1	21	1h 17m 36s
base-front/filler-corner-L.stl	1	21	2h 0m 50s
base-front/filler-corner-R.stl	1	21	1h 56m 31s
base-rear/former-split-gasket-groove-part1of4.stl	1	2	9m 50s
base-rear/former-split-gasket-groove-part2of4.stl	1	1	8m 58s
base-rear/former-split-gasket-groove-part3of4.stl	1	1	8m 58s
base-rear/former-split-gasket-groove-part4of4.stl	1	2	9m 53s
base-front/core-center-post.stl	1	5	44m 56s
base-rear/core-center-post.stl	1	5	44m 56s
base-front/core-outboard-L-post.stl	1	5	44m 53s
base-front/core-trough-band-L-post.stl	1	5	44m 54s
base-front/core-draw-rod-L-post.stl	1	5	44m 53s
base-front/core-foot-a-L-post.stl	1	5	44m 53s
base-rear/core-foot-b-L-post.stl	1	5	44m 53s
base-rear/core-knee-L-post.stl	1	5	44m 53s
base-rear/core-outer-a-L-post.stl	1	5	44m 53s
base-rear/core-outer-b-L-post.stl	1	5	44m 53s
base-rear/core-rear-L-post.stl	1	5	44m 53s
base-front/core-outboard-R-post.stl	1	5	44m 53s
base-front/core-trough-band-R-post.stl	1	5	44m 54s
base-front/core-draw-rod-R-post.stl	1	5	44m 53s
base-front/core-foot-a-R-post.stl	1	5	44m 53s
base-rear/core-foot-b-R-post.stl	1	5	44m 53s
base-rear/core-knee-R-post.stl	1	5	44m 53s
base-rear/core-outer-a-R-post.stl	1	5	44m 53s
base-rear/core-outer-b-R-post.stl	1	5	44m 53s
base-rear/core-rear-R-post.stl	1	5	44m 53s
base-rear/plug-socket-spike.stl	4	8	13m 26s
base-rear/kit-spigot-bushing.stl	2	34	1h 11m 37s
upright/kit-back-bracket.stl	4	21	27m 33s
upright/kit-front-bracket.stl	4	32	38m 27s
upright/kit-saddle-flat.stl	4	18	22m 40s

File	Qty	g	Time each
`upright/filler-toe-filler.stl`	2	54	1h 33m 58s
`upright/former-foot-gasket-groove-part1of4.stl`	2	7	17m 8s
`upright/former-foot-gasket-groove-part2of4.stl`	2	3	8m 46s
`upright/former-foot-gasket-groove-part3of4.stl`	2	3	8m 46s
`upright/former-foot-gasket-groove-part4of4.stl`	2	7	17m 10s
`upright/plug-slope-plug-lower.stl`	2	29	1h 0m 39s
`upright/plug-slope-plug-upper.stl`	2	29	1h 0m 39s
`upright/kit-plate-spacer.stl`	4	15	20m 9s
`upright/kit-post-bushing.stl`	4	67	1h 11m 51s
`beam/kit-corner-bracket.stl`	4	33	40m 3s
`beam/filler-back-edge-part1of4.stl`	1	19	1h 14m 46s
`beam/filler-back-edge-part2of4.stl`	1	19	1h 14m 46s
`beam/filler-back-edge-part3of4.stl`	1	19	1h 14m 46s
`beam/filler-back-edge-part4of4.stl`	1	19	1h 15m 1s
`beam/kit-tube-locator.stl`	4	43	58m 26s
`beam/kit-dowel-stop.stl`	2	1	4m 19s
`tools/kit-pour-gauge.stl`	2	9	24m 33s
`tools/kit-screed-shoe.stl`	2	20	44m 42s
`base-front/cast-in-base-front-tube-plug-l.stl`	2	15	39m 34s
`base-front/cast-in-base-front-spigot-sleeve-l.stl`	2	22	59m 28s
`base-front/cast-in-leg-front-l-plug.stl`	2	8	24m 14s
`base-rear/cast-in-base-rear-tube-plug-l.stl`	2	15	39m 16s
`base-rear/cast-in-base-rear-spigot-sleeve-l.stl`	2	22	59m 31s
`base-rear/cast-in-leg-rear-l-plug.stl`	2	8	24m 14s
`upright/cast-in-post-l1-plug.stl`	4	16	24m 37s
`upright/cast-in-rod-sleeve-l1a.stl`	8	61	57m 53s

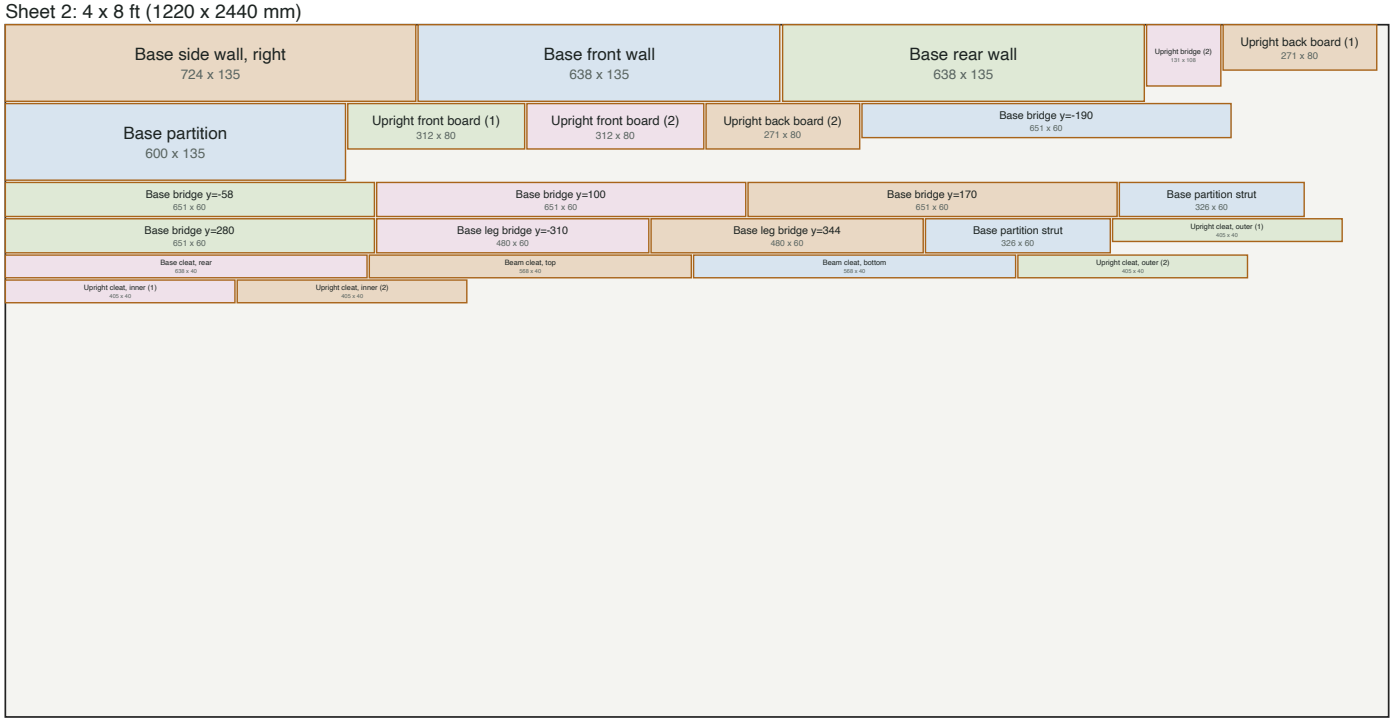
4. Cut the boards

Have the store rip the strips; crosscut at home. Write each board's name on its outside face, and mark the cast face (the melamine face the mix touches).

Sheet 1: 4 x 8 ft (1220 x 2440 mm)



Sheet 2: 4 x 8 ft (1220 x 2440 mm)



Board	Qty	Size (mm)	Notes
Base floor	1	718 x 843	both halves are cast on it
Base front wall	1	638 x 135	
Base rear wall	1	638 x 135	
Base side wall, left	1	724 x 135	
Base side wall, right	1	724 x 135	

Board	Qty	Size (mm)	Notes
Base partition	1	600 x 135	rear half only; comes out before the front half
Base partition strut	2	326 x 60	offcut, stands on edge; holds the partition foot (rear half only)
Base cleat, front	1	638 x 40	offcut strip, flat on the floor
Base cleat, rear	1	638 x 40	offcut strip, flat on the floor
Base cleat, left	1	724 x 40	offcut strip, flat on the floor
Base cleat, right	1	724 x 40	offcut strip, flat on the floor
Base bridge y=-190	1	651 x 60	
Base bridge y=-58	1	651 x 60	
Base bridge y=100	1	651 x 60	
Base bridge y=170	1	651 x 60	
Base bridge y=280	1	651 x 60	
Base leg bridge y=-310	1	480 x 60	
Base leg bridge y=344	1	480 x 60	
Upright floor	2	198 x 575	
Upright side plate, outer	2	495 x 271 (profile)	cut to the profile
Upright side plate, inner	2	495 x 271 (profile)	cut to the profile
Upright back board	2	80 x 271	
Upright front board	2	80 x 312	ends cut square
Upright bridge	2	131 x 108	
Upright cleat, outer	2	405 x 40	offcut strip, flat on the floor
Upright cleat, inner	2	405 x 40	offcut strip, flat on the floor
Upright cleat, back	2	80 x 40	offcut strip, flat on the floor
Beam floor	1	648 x 258	
Beam top wall	1	568 x 165	
Beam bottom wall	1	568 x 165	
Beam end wall, left	1	140 x 165	
Beam end wall, right	1	140 x 165	
Beam cleat, top	1	568 x 40	offcut strip, flat on the floor
Beam cleat, bottom	1	568 x 40	offcut strip, flat on the floor
Beam cleat, left	1	140 x 40	offcut strip, flat on the floor
Beam cleat, right	1	140 x 40	offcut strip, flat on the floor

5. Mark and drill

1. Print the drilling templates (`templates/base-templates.pdf`, `templates/upright-templates.pdf`, `templates/beam-templates.pdf`) at **actual size** and check the 100 mm square.

2. Tape each board's pages onto it: printed edges on the board's edges, grid lines lined up.
3. Awl through every crosshair. Pencil the dashed **pour line** on the walls.
4. Drill from the marked face, at the size the color legend gives. Pilot the big ones; the 63.5 mm holes are a 2-1/2 in hole saw.

The appendix has every board as a drawing and a table, if you would rather measure.

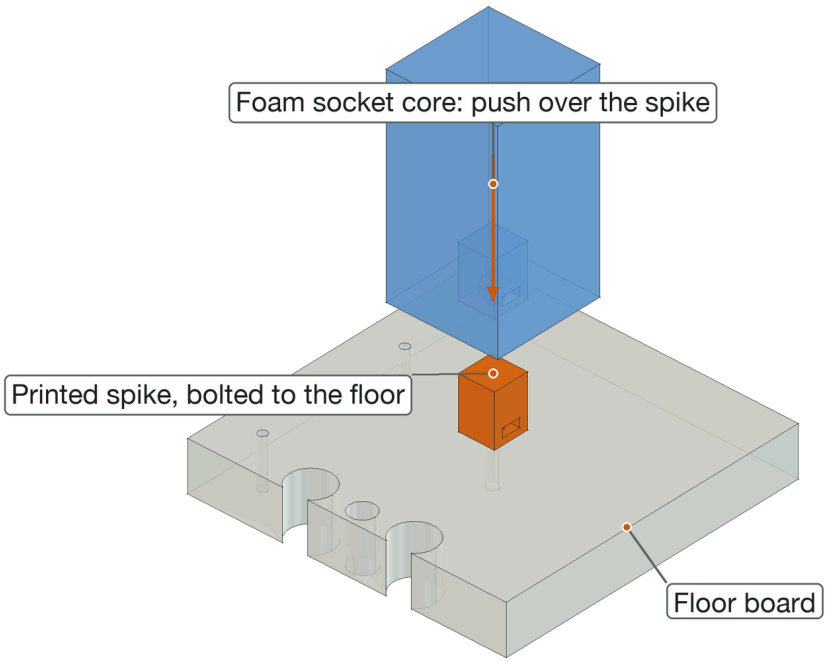
Hole	Drill	For
3.2 mm	1/8 in	foot-gasket-groove, motor-relief, split-gasket-groove
3.4 mm	9/64 in	corner-L, corner-R, gutter-L, gutter-R
4.5 mm	11/64 in	back-edge, bar hanger, cleat, core post, corner bracket, corner-L, corner-R, gutter-L, gutter-R, leg holder, saddle, slope-plug-lower, slope-plug-upper, socket spike, toe-filler, trough, tube locator
5.5 mm	7/32 in	M5 insert screw
6.5 mm	17/64 in	M6 insert screw
9.6 mm	25/64 in	3/8 dowel
10 mm	25/64 in	3/8-16 bolt into the top plate
13.1 mm	33/64 in	1/2 core rod, 1/2-13 bolt, base dowel hole
13.5 mm	17/32 in	1/2 rod through a sleeve, 1/2-13 bolt into coupler
16.3 mm	41/64 in	5/8 core rod
63.5 mm	2-1/2 in hole saw	hole saw

6. Cut the foam

Fine saw, hot wire or a long knife. Stack and glue two layers of 2 in foam for the 90 mm cores. Wrap each piece smooth in packing tape. Cut a snug 12 mm square hole 30 mm deep for each core's post and glue the post in.

Casting	Piece	Qty	Size and how
base-front	center	1	XPS block 100 x 185 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	center	1	XPS block 100 x 190 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	outboard-L	1	XPS block 130 x 164 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	trough-band-L	1	XPS block 101 x 55 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.

Casting	Piece	Qty	Size and how
base-front	draw-rod-L	1	XPS block 20 x 255 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	foot-a-L	1	XPS block 45 x 66 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-b-L	1	XPS block 45 x 99 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	knee-L	1	XPS block 50 x 184 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-a-L	1	XPS block 20 x 44 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-b-L	1	XPS block 20 x 60 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	rear-L	1	XPS block 170 x 52 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	outboard-R	1	XPS block 130 x 164 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	trough-band-R	1	XPS block 101 x 55 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	draw-rod-R	1	XPS block 20 x 255 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-front	foot-a-R	1	XPS block 45 x 66 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	foot-b-R	1	XPS block 45 x 99 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	knee-R	1	XPS block 50 x 184 x 40 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-a-R	1	XPS block 20 x 44 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	outer-b-R	1	XPS block 20 x 60 x 65 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	rear-R	1	XPS block 170 x 52 x 90 mm, outline as the pocket (chamfered corners); taped, with its printed post glued in.
base-rear	socket	4	XPS block 43 x 43 x 80 mm, taped; dig it out after curing (a little acetone dissolves the rest).



7. Wax

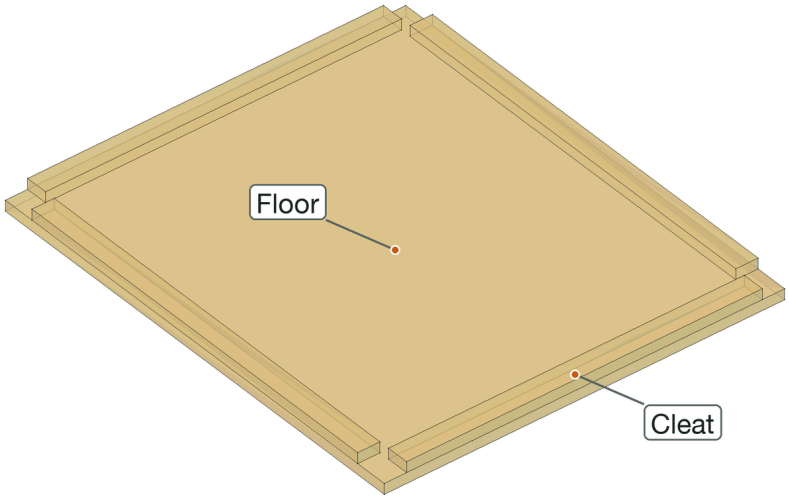
- 3-5 coats of paste wax on every face the mix touches, boards and removable printed pieces, then PVA. Don't wax pieces that stay in (core posts, cast-in parts): scuff and degrease them.
- Glue quarter-round into the floor-to-wall corners that make outside edges; silicone fillets elsewhere.
- Grease every screw, bolt and rod that comes out of the casting.

8. Assemble

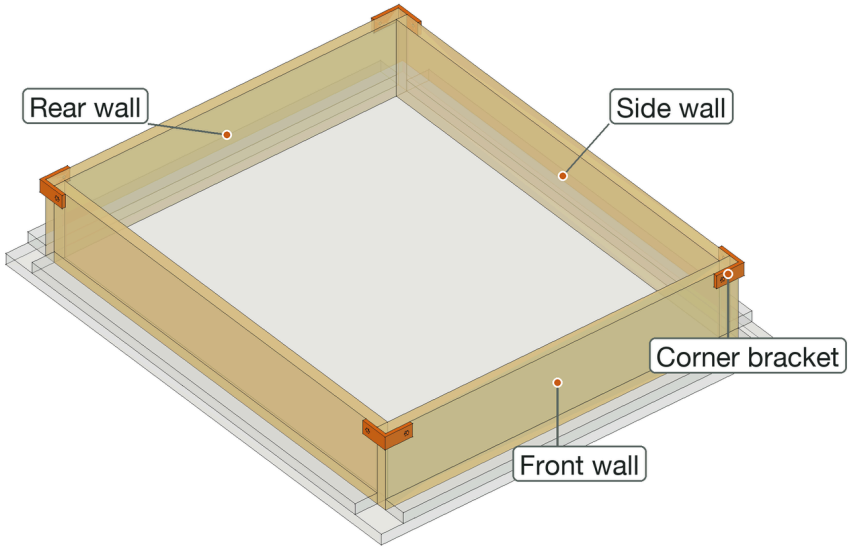
Base, rear half (first)

Cast upside down, top face on the floor; shown the right way up. New parts in color.

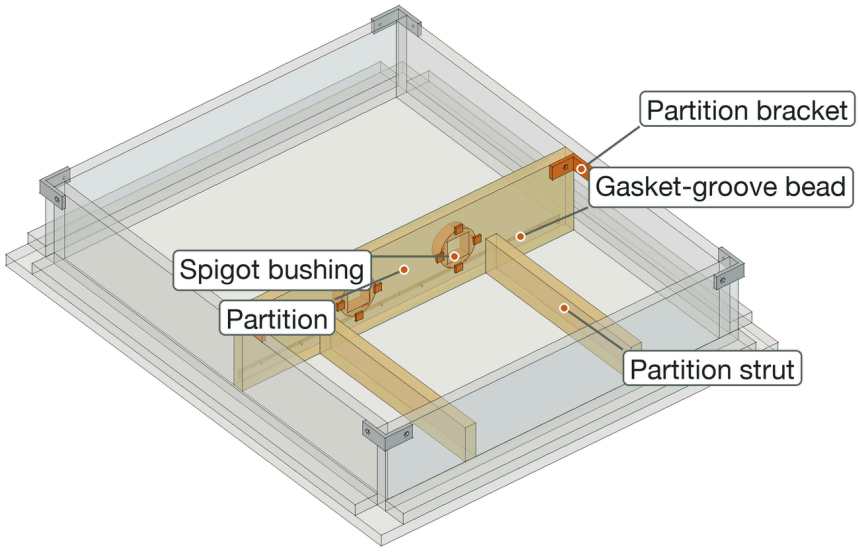
1. Floor on three battens. Bolt the cleats down.



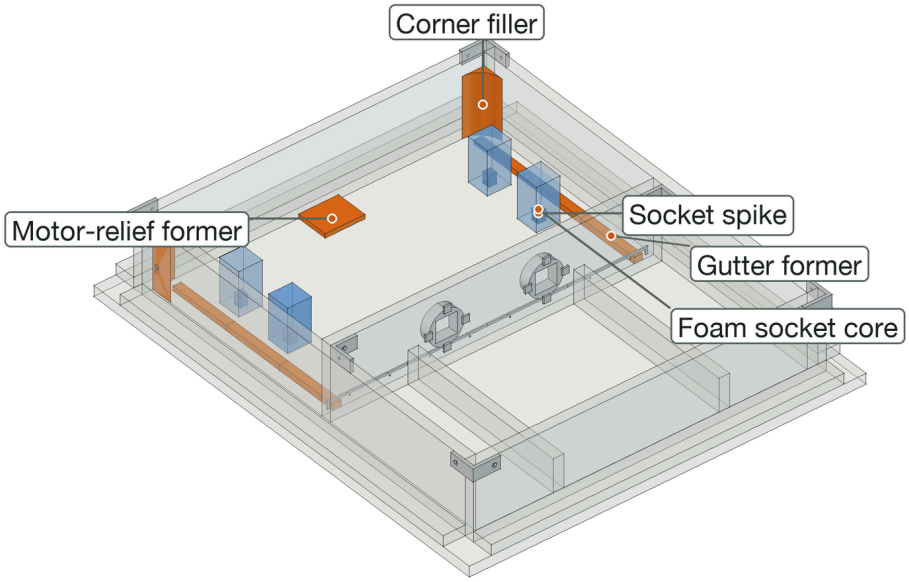
2. Walls against the cleats; corner brackets at the top. Diagonals equal within 2 mm, then tighten.



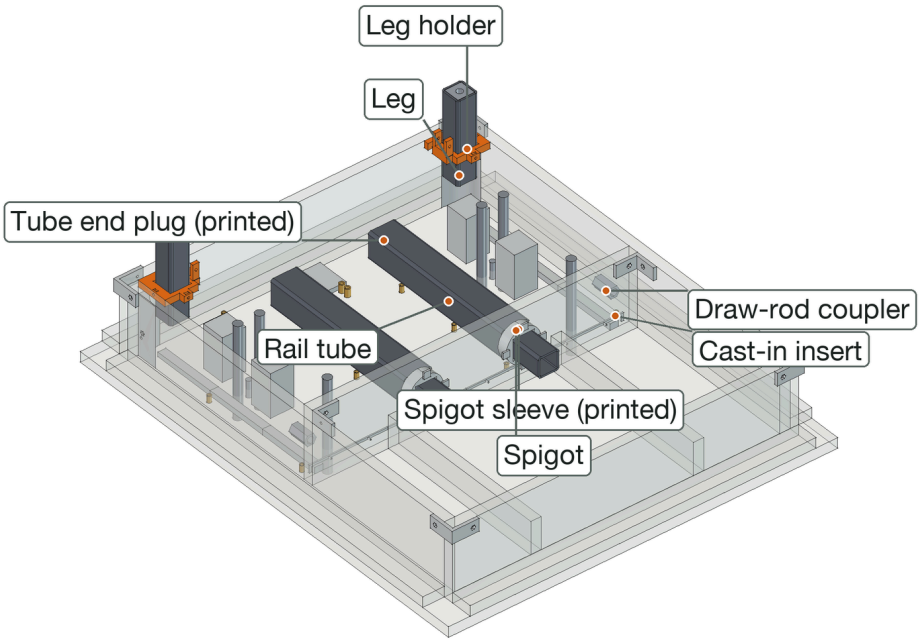
3. Partition between the side walls: two brackets, two struts to the front wall. Beads on its pins, spigot bushings in its holes.



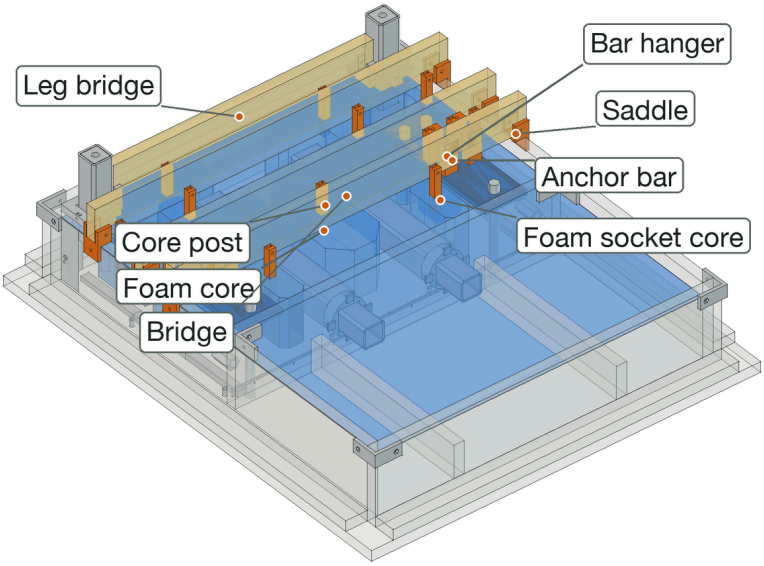
4. Gutter formers, motor relief and corner fillers on; socket spikes bolted down with a foam block on each.



5. Inserts on their greased screws; greased 5/8 in rods up through the floor for the tie-rod holes; spigots through the bushings to their middle marks, rear tubes on them, split ends tight to the partition; leg holders in the corners, legs set to height, pinch screws tight.

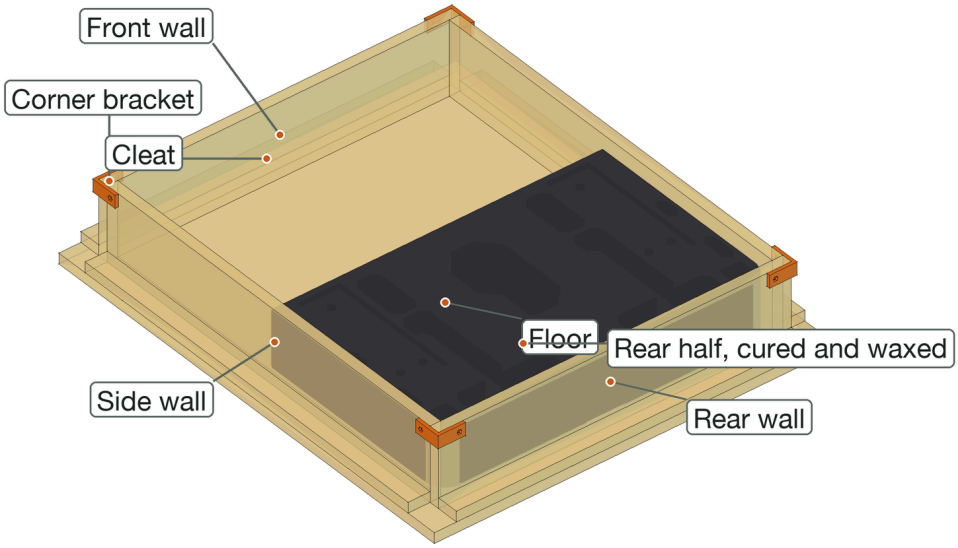


6. Pour the 20 mm skin. Foam cores on it, bridges into the saddles, each post bolted to its bridge. Fill to the pour line; the anchor bars over the rods, screwed to their hangers, flush with the line.

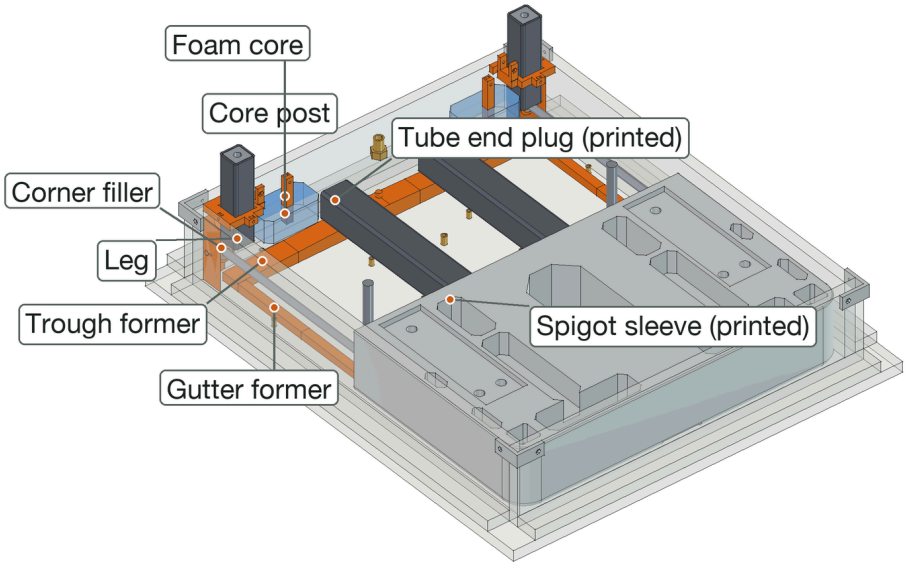


Base, front half (against the rear half)

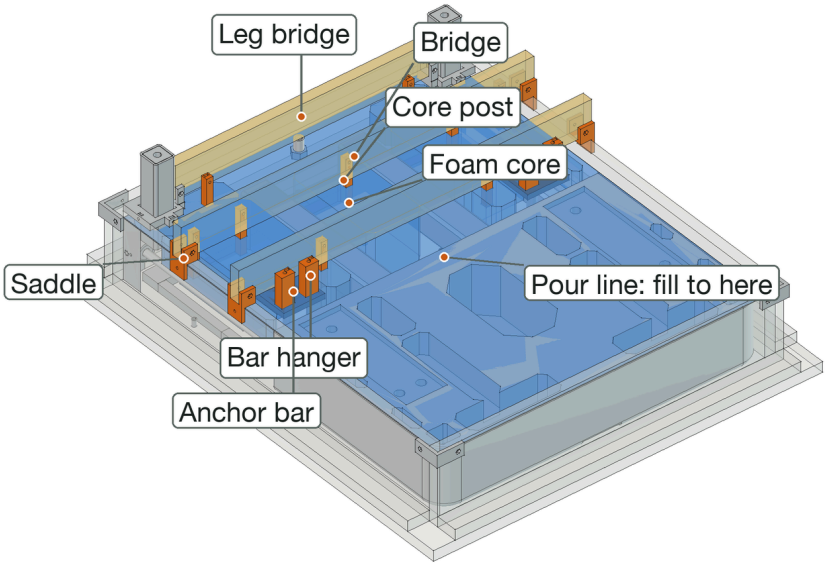
1. After the cure: bridges, partition and struts off. Dig out the socket foam, pull the beads, tape the groove, and wax the rear half's split face and spigots.



2. Trough and gutter formers and corner fillers on; front tubes onto the spigots; draw rods into the rear couplers; inserts, tie-rod core rods and legs.



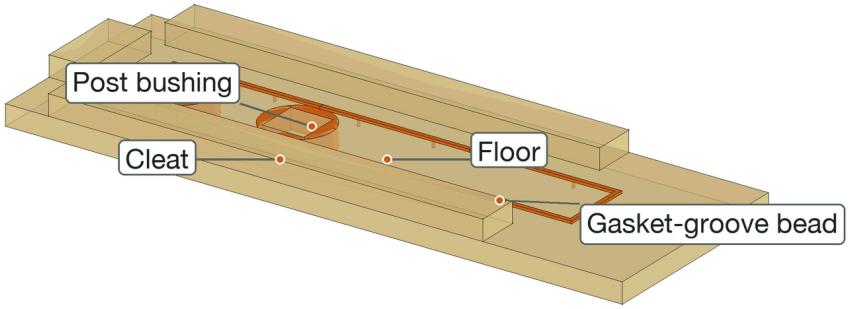
3. Skin, cores, bridges, fill; the anchor bars on their hangers. After the cure, pull the rods, take it apart and split the halves with wedges.



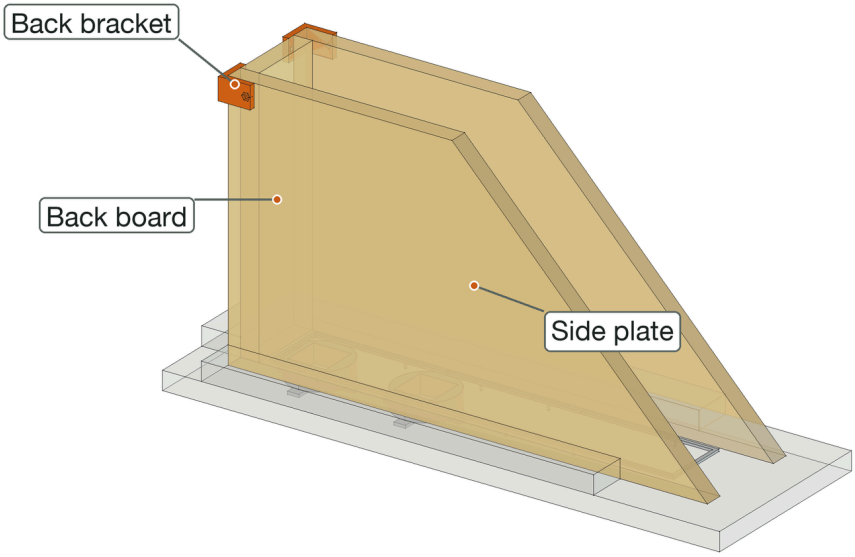
Uprights (two molds)

Cast foot down. Build two of everything.

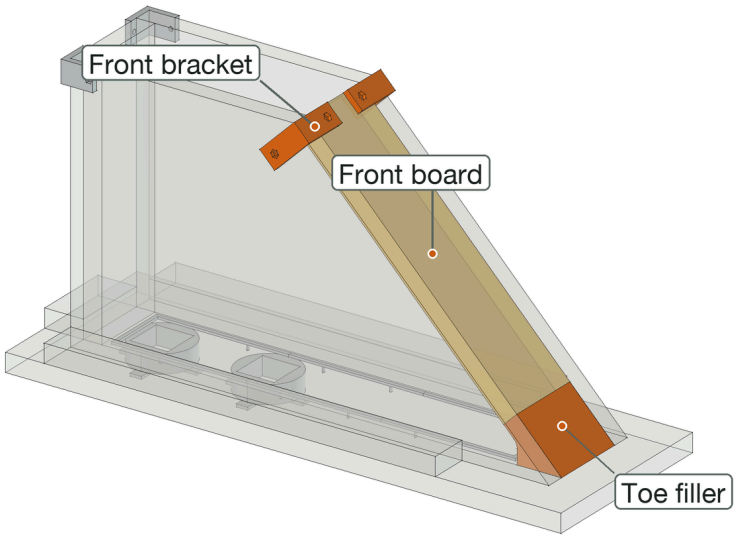
1. Post bushings into the floor from below; gasket bead pinned on; cleats bolted down.



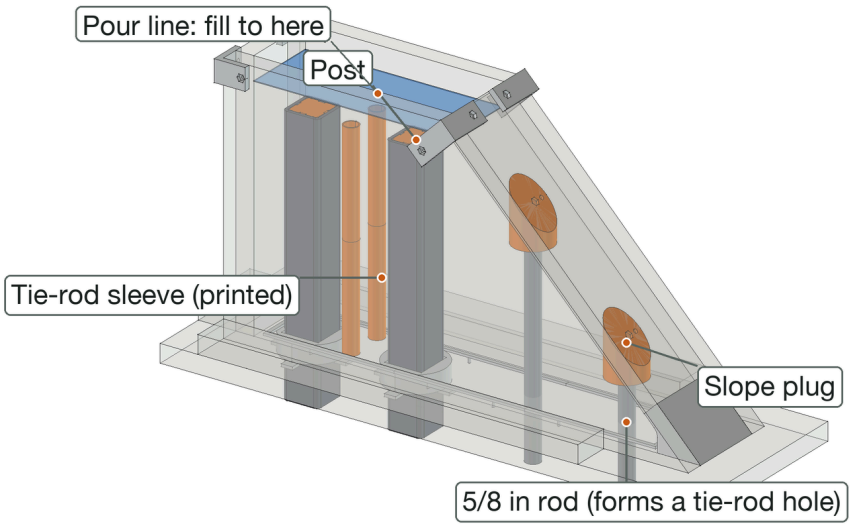
2. Side plates against their cleats, back board between them, two back brackets.



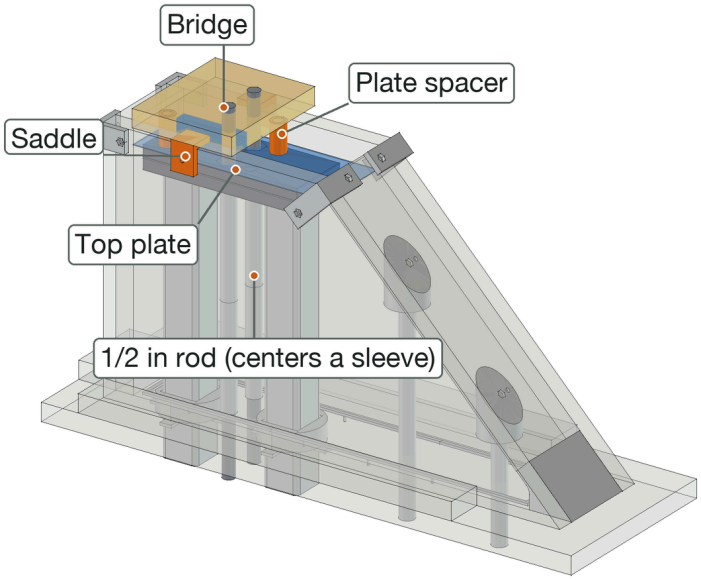
3. Toe filler bolted down, the front board's foot in its lip, two front brackets at the top.

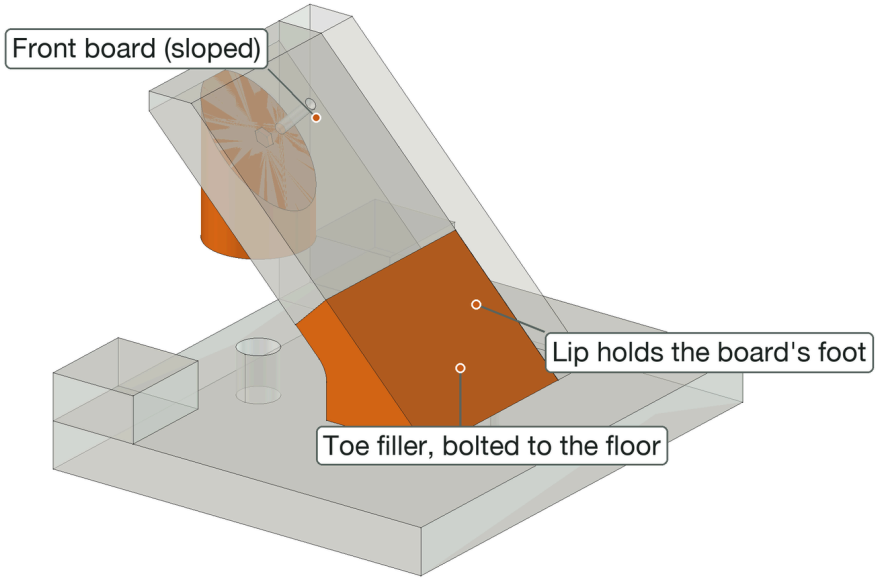


4. Mold on its four 2x4 blocks. Posts up through the bushings, standing on the bench (a long post just ends nearer the top plate: 6 mm room); greased 5/8 in rods up through the floor into the slope plugs; the two top sleeves standing over their floor holes, tops taped; dowel-hole bolt from underneath.



5. Fill to the line. Top plate (both layers) hung from the bridge, bridge onto its saddles; greased 1/2 in rods up from under the floor, through the sleeves, the plate and the bridge.

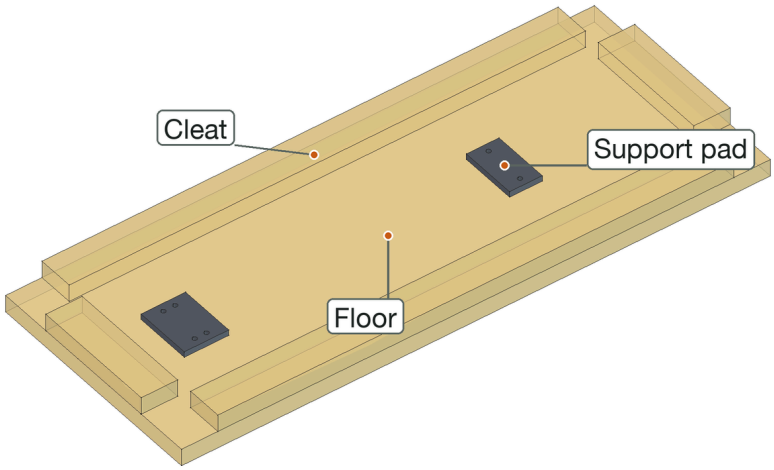




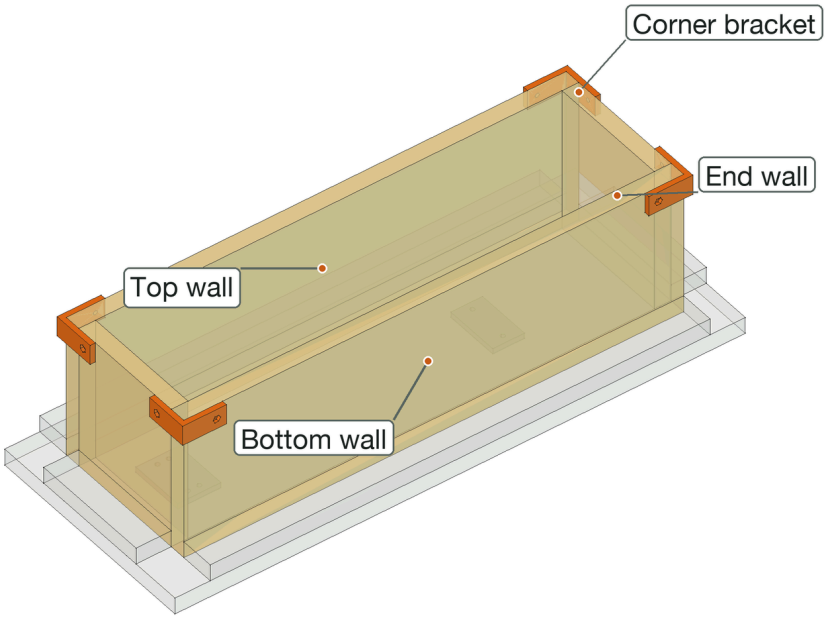
Beam

Cast front (rail) face down, back open.

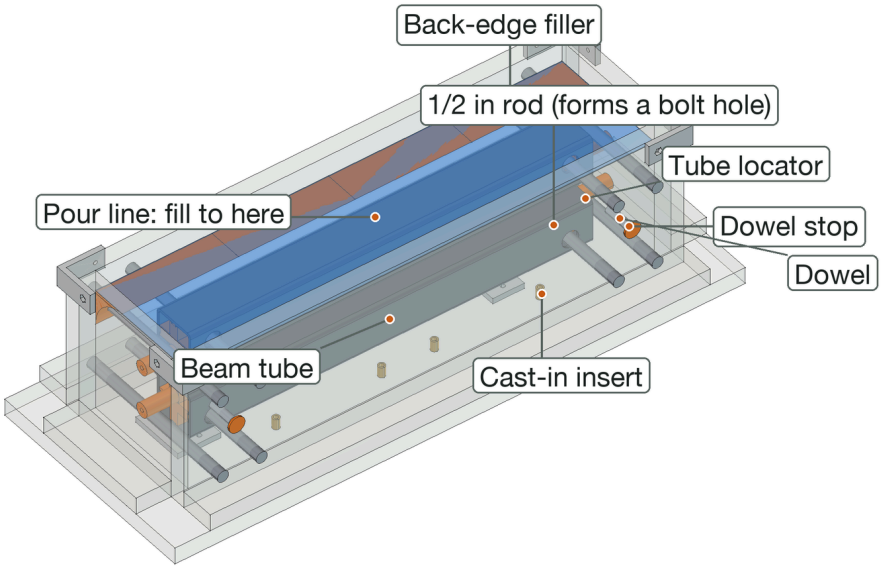
1. Cleats bolted down; support pads and riser inserts screwed to the floor.

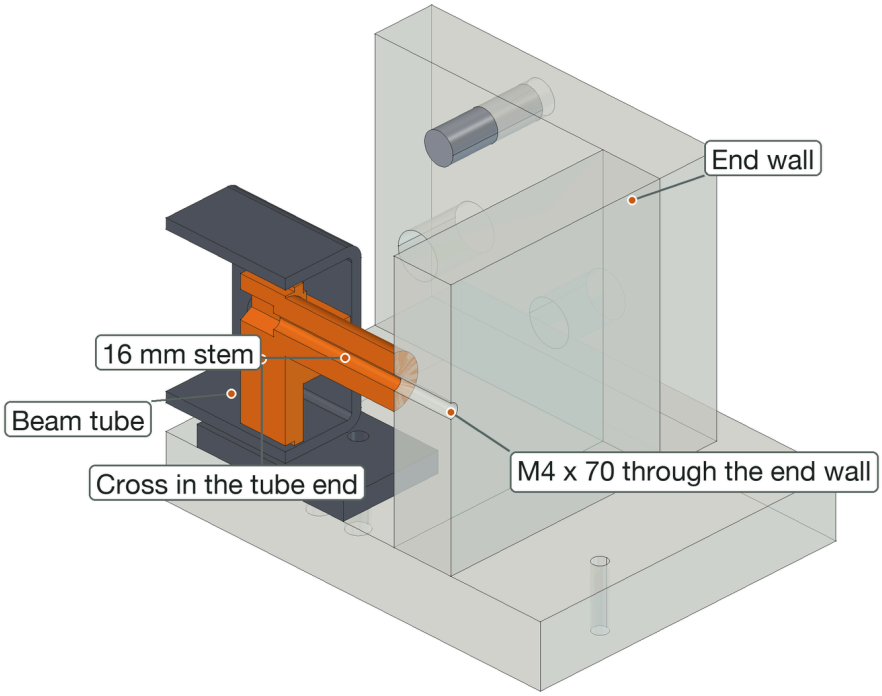


2. Walls against the cleats; four corner brackets.



3. Tube locators bolted through the end walls. Butter the support pads with thickened epoxy (about 3 mm), then the tubes between the locators, holes over the rod holes; back-edge filler on the top wall; dowels on their stops; greased 1/2 in rods through both walls and the tubes. Fill to the line.

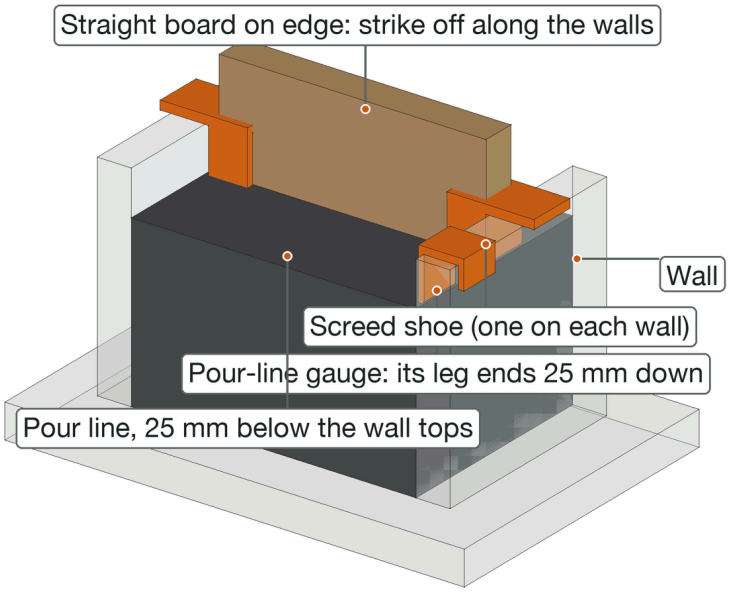




9. Pour

- Fill to the **pour line**, 25 mm below the wall tops (pencilled from the templates, or with the printed pour-line gauge clipped over a wall).
- Vibrate each layer until it stops bubbling; check the brackets halfway through.

- Strike the top off with a straight board in the two printed **screed shoes**: they hang it from the wall tops with its bottom edge on the pour line.



10. Take apart

1. After the full cure: bridges and saddles off, then corner brackets, then cleats; pull the walls off.
2. Screws, bolts and rods out from outside; lift the casting off the floor.
3. Unbolt and pull the formers and fillers. Dig out the socket foam (a little acetone helps).
4. Scrape and re-wax the boards; keep each mold's kit in a box. The kit is reused.

11. If something doesn't fit

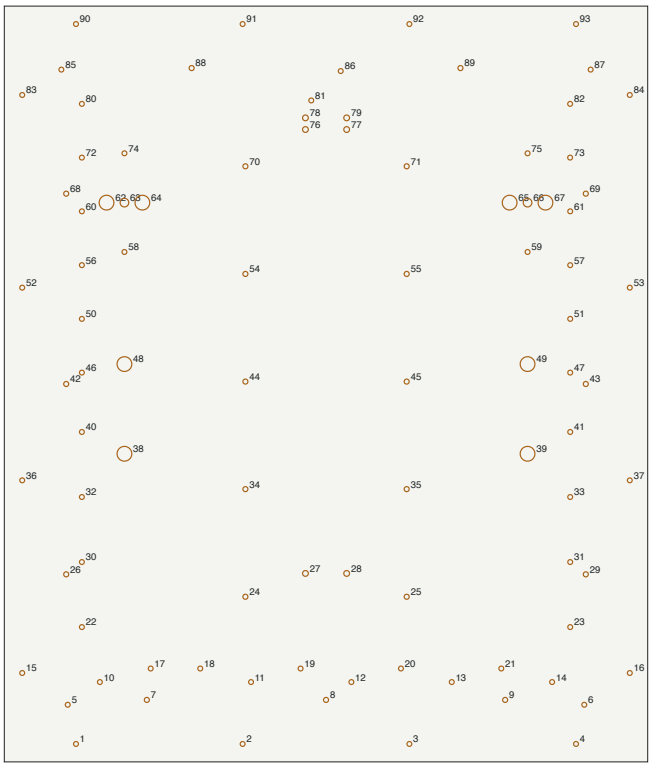
Problem	Fix
Nut won't slide under the hole	Clean the pocket with a nut on a bolt; warm the nut
Bracket or saddle loose on the board	Board thinner than set: tape shim, or set `mold_board` and reprint
Pin won't go in	Run the 1/8 in bit through again; trim the pin tip
Foam core moves	Tighten its post bolt; carpet tape under it
Printed part won't go on or into a steel tube	File the weld seam; sand the part. Several tight: caliper the tube, put its sizes in `frame_tube` / `post_tube`, re-run and reprint
Printed part loose on a tube	A wrap of tape round the part
Leak at a corner	Silicone fillet inside before waxing; check the brackets and diagonals
Former stuck after curing	Bolts out first, then a dowel and a light tap

Appendix: hole maps

Every board as seen from the face you mark: x from the left edge, y from the bottom edge (for a wall, the edge on the floor). The templates carry the same holes at full size.

Base

Base floor



0,0 · 718 x 843 mm · seen from the cast side

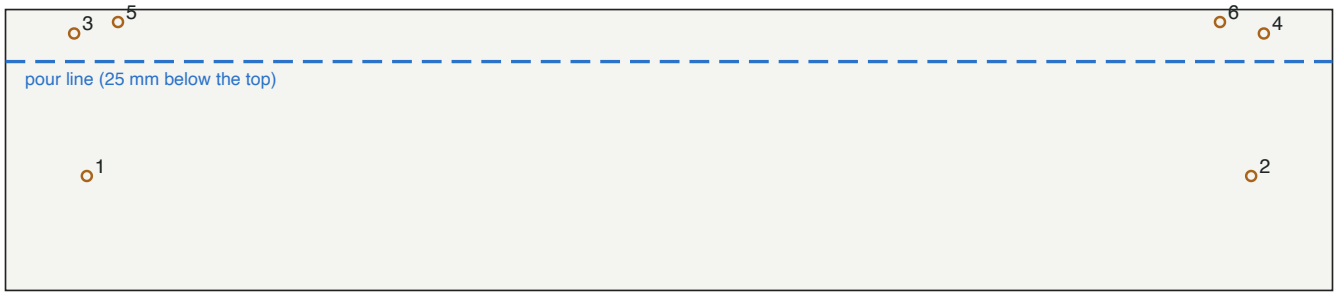
#	x	y	Hole	For
1	80	20	4.5	cleat
2	266	20	4.5	cleat
3	452	20	4.5	cleat
4	638	20	4.5	cleat
5	71	64	3.4	corner-R
6	647	64	3.4	corner-L
7	159	69	5.5	M5 insert screw
8	359	69	5.5	M5 insert screw
9	559	69	5.5	M5 insert screw
10	107	89	4.5	trough
11	275	89	4.5	trough
12	387	89	4.5	trough

#	x	y	Hole	For
13	499	89	4.5	trough
14	611	89	4.5	trough
15	20	99	4.5	cleat
16	698	99	4.5	cleat
17	163	104	4.5	trough
18	219	104	4.5	trough
19	331	104	4.5	trough
20	443	104	4.5	trough
21	555	104	4.5	trough
22	87	150	4.5	gutter-R
23	632	150	4.5	gutter-L
24	269	184	5.5	M5 insert screw
25	449	184	5.5	M5 insert screw
26	69	209	5.5	M5 insert screw
27	336	210	6.5	M6 insert screw
28	382	210	6.5	M6 insert screw
29	649	209	5.5	M5 insert screw
30	87	223	4.5	gutter-R
31	632	223	4.5	gutter-L
32	87	295	3.4	gutter-R
33	632	295	3.4	gutter-L
34	269	304	5.5	M5 insert screw
35	449	304	5.5	M5 insert screw
36	20	314	4.5	cleat
37	698	314	4.5	cleat
38	134	344	16.3	5/8 core rod
39	584	344	16.3	5/8 core rod
40	87	368	3.4	gutter-R
41	632	368	3.4	gutter-L
42	69	421	5.5	M5 insert screw
43	649	421	5.5	M5 insert screw
44	269	424	5.5	M5 insert screw
45	449	424	5.5	M5 insert screw
46	87	434	3.4	gutter-R
47	632	434	3.4	gutter-L

#	x	y	Hole	For
48	134	444	16.3	5/8 core rod
49	584	444	16.3	5/8 core rod
50	87	494	3.4	gutter-R
51	632	494	3.4	gutter-L
52	20	529	4.5	cleat
53	698	529	4.5	cleat
54	269	544	5.5	M5 insert screw
55	449	544	5.5	M5 insert screw
56	87	554	3.4	gutter-R
57	632	554	3.4	gutter-L
58	134	569	4.5	socket spike
59	584	569	4.5	socket spike
60	87	614	3.4	gutter-R
61	632	614	3.4	gutter-L
62	114	624	16.3	5/8 core rod
63	134	624	9.6	3/8 dowel
64	154	624	16.3	5/8 core rod
65	564	624	16.3	5/8 core rod
66	584	624	9.6	3/8 dowel
67	604	624	16.3	5/8 core rod
68	69	634	5.5	M5 insert screw
69	649	634	5.5	M5 insert screw
70	269	664	5.5	M5 insert screw
71	449	664	5.5	M5 insert screw
72	87	674	3.4	gutter-R
73	632	674	3.4	gutter-L
74	134	679	4.5	socket spike
75	584	679	4.5	socket spike
76	336	705	6.5	M6 insert screw
77	382	705	6.5	M6 insert screw
78	336	718	6.5	M6 insert screw
79	382	718	6.5	M6 insert screw
80	87	734	3.4	gutter-R
81	343	737	3.2	motor-relief
82	632	734	3.4	gutter-L

#	x	y	Hole	For
83	20	744	4.5	cleat
84	698	744	4.5	cleat
85	64	772	3.4	corner-R
86	375	770	3.2	motor-relief
87	654	772	3.4	corner-L
88	209	774	5.5	M5 insert screw
89	509	774	5.5	M5 insert screw
90	80	823	4.5	cleat
91	266	823	4.5	cleat
92	452	823	4.5	cleat
93	638	823	4.5	cleat

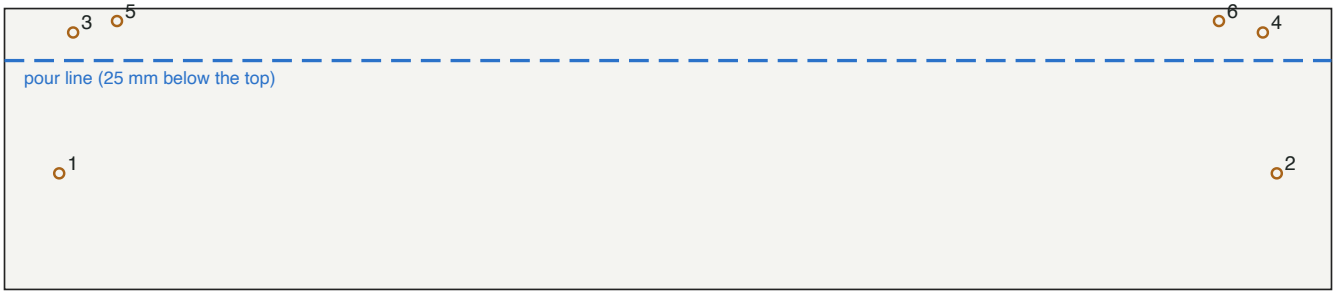
Base front wall



0,0 · 638 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	39	55	4.5	corner-L
2	599	55	4.5	corner-R
3	33	124	4.5	corner bracket
4	605	124	4.5	corner bracket
5	54	129	4.5	leg holder
6	584	129	4.5	leg holder

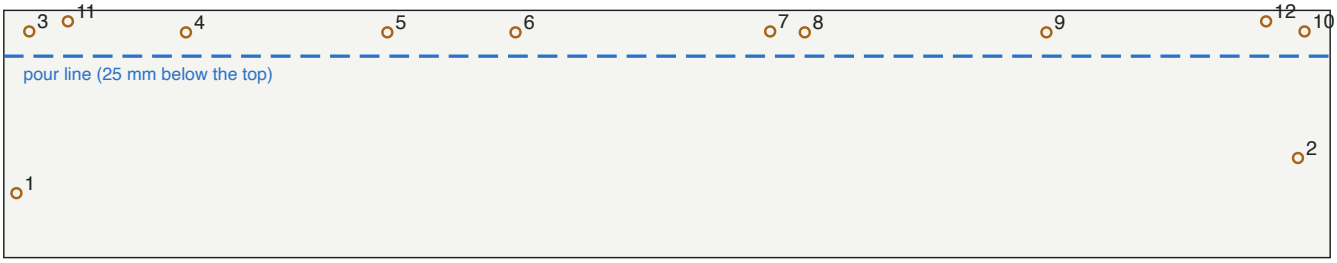
Base rear wall



0,0 · 638 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	26	56	4.5	corner-R
2	612	56	4.5	corner-L
3	33	124	4.5	corner bracket
4	605	124	4.5	corner bracket
5	54	129	4.5	leg holder
6	584	129	4.5	leg holder

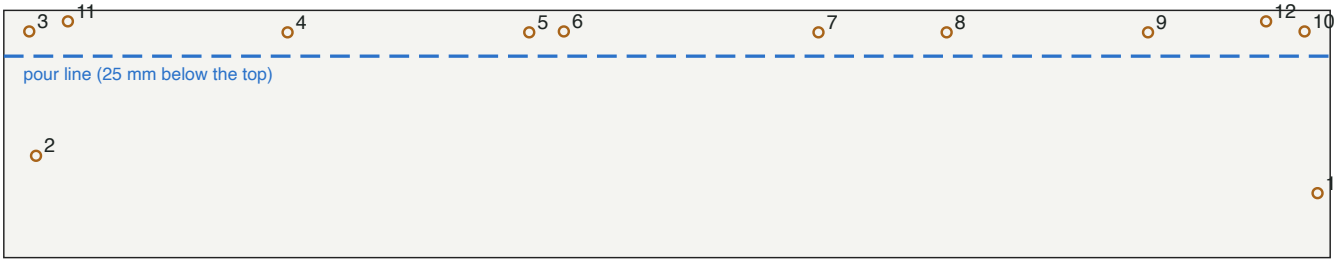
Base side wall, left



0,0 · 724 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	7	35	4.5	corner-L
2	707	54	4.5	corner-L
3	14	124	4.5	corner bracket
4	100	123	4.5	saddle
5	210	123	4.5	saddle
6	280	123	4.5	saddle
7	419	124	4.5	corner bracket
8	438	123	4.5	saddle
9	570	123	4.5	saddle
10	711	124	4.5	corner bracket
11	35	129	4.5	leg holder
12	690	129	4.5	leg holder

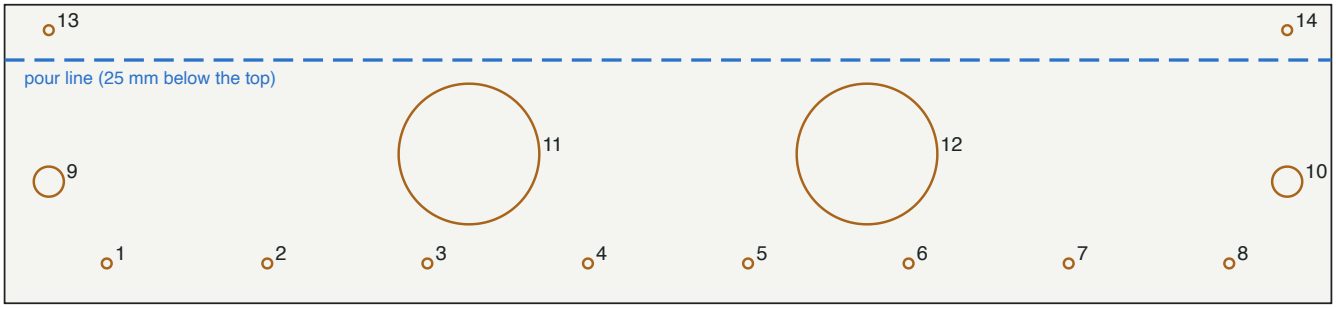
Base side wall, right



0,0 · 724 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	718	35	4.5	corner-R
2	18	56	4.5	corner-R
3	14	124	4.5	corner bracket
4	155	123	4.5	saddle
5	287	123	4.5	saddle
6	306	124	4.5	corner bracket
7	445	123	4.5	saddle
8	515	123	4.5	saddle
9	625	123	4.5	saddle
10	711	124	4.5	corner bracket
11	35	129	4.5	leg holder
12	690	129	4.5	leg holder

Base partition



0,0 · 600 x 135 mm · seen from the cast side

#	x	y	Hole	For
1	46	18	3.2	split-gasket-groove
2	119	18	3.2	split-gasket-groove
3	191	18	3.2	split-gasket-groove
4	264	18	3.2	split-gasket-groove

#	x	y	Hole	For
5	336	18	3.2	split-gasket-groove
6	409	18	3.2	split-gasket-groove
7	481	18	3.2	split-gasket-groove
8	554	18	3.2	split-gasket-groove
9	20	55	13.5	1/2-13 bolt into coupler
10	580	55	13.5	1/2-13 bolt into coupler
11	210	68	63.5	hole saw
12	390	68	63.5	hole saw
13	20	124	4.5	corner bracket
14	580	124	4.5	corner bracket

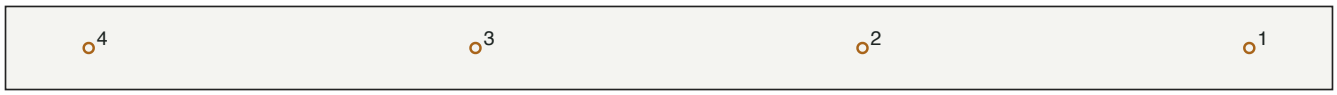
Base cleat, front



0,0 · 638 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	598	20	4.5	cleat
2	412	20	4.5	cleat
3	226	20	4.5	cleat
4	40	20	4.5	cleat

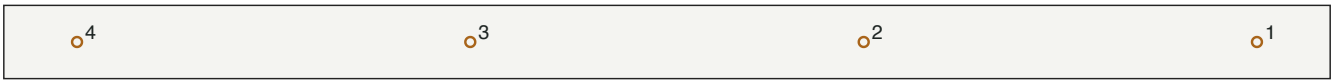
Base cleat, rear



0,0 · 638 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	598	20	4.5	cleat
2	412	20	4.5	cleat
3	226	20	4.5	cleat
4	40	20	4.5	cleat

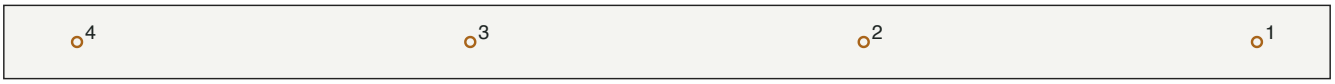
Base cleat, left



0,0 · 724 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	684	20	4.5	cleat
2	470	20	4.5	cleat
3	255	20	4.5	cleat
4	40	20	4.5	cleat

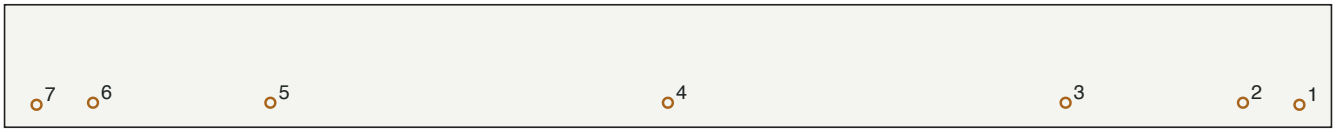
Base cleat, right



0,0 · 724 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	684	20	4.5	cleat
2	470	20	4.5	cleat
3	255	20	4.5	cleat
4	40	20	4.5	cleat

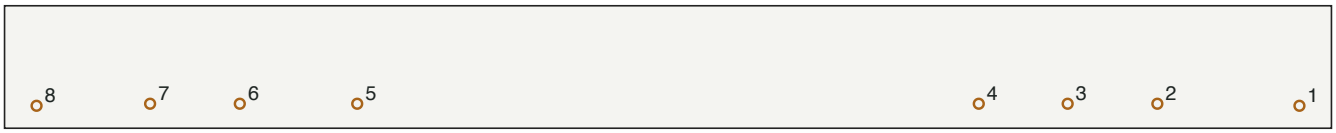
Base bridge y=-190



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	608	12	4.5	core post
3	520	12	4.5	core post
4	325	12	4.5	core post
5	130	12	4.5	core post
6	43	12	4.5	core post
7	16	11	4.5	saddle

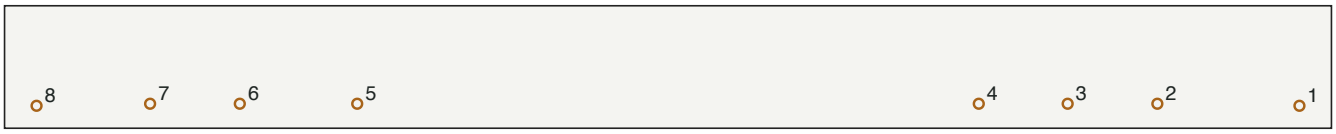
Base bridge y=-58



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	565	12	4.5	bar hanger
3	521	12	4.5	bar hanger
4	478	12	4.5	core post
5	173	12	4.5	core post
6	115	12	4.5	bar hanger
7	71	12	4.5	bar hanger
8	16	11	4.5	saddle

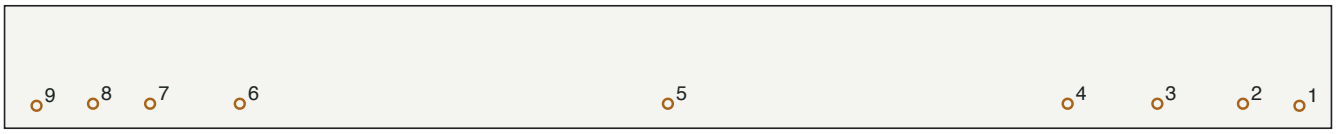
Base bridge y=100



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	565	12	4.5	bar hanger
3	521	12	4.5	bar hanger
4	478	12	4.5	core post
5	173	12	4.5	core post
6	115	12	4.5	bar hanger
7	71	12	4.5	bar hanger
8	16	11	4.5	saddle

Base bridge y=170



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	608	12	4.5	core post
3	565	12	4.5	bar hanger
4	521	12	4.5	bar hanger
5	325	12	4.5	core post
6	115	12	4.5	bar hanger
7	71	12	4.5	bar hanger
8	43	12	4.5	core post
9	16	11	4.5	saddle

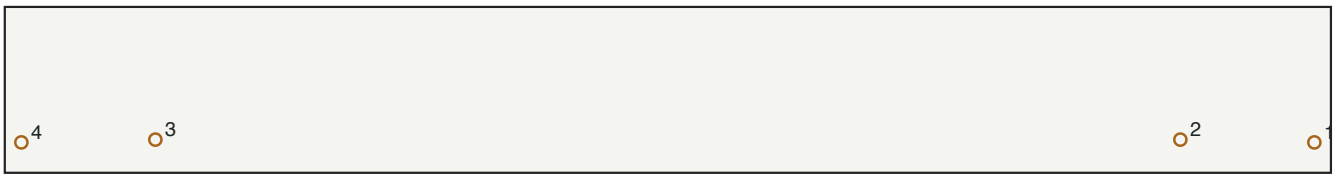
Base bridge y=280



0,0 · 651 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	635	11	4.5	saddle
2	608	12	4.5	core post
3	480	12	4.5	core post
4	170	12	4.5	core post
5	43	12	4.5	core post
6	16	11	4.5	saddle

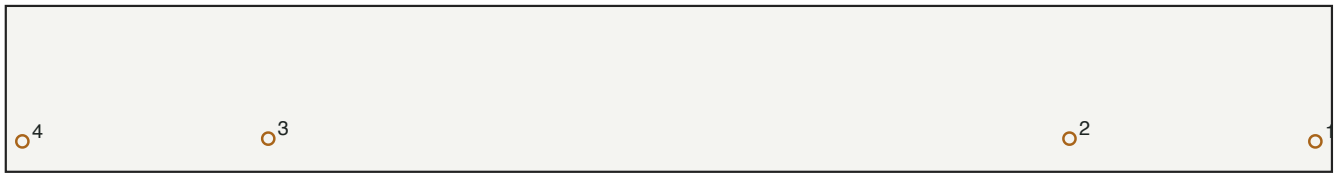
Base leg bridge y=-310



0,0 · 480 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	474	11	4.5	leg holder
2	426	12	4.5	core post
3	54	12	4.5	core post
4	6	11	4.5	leg holder

Base leg bridge y=344

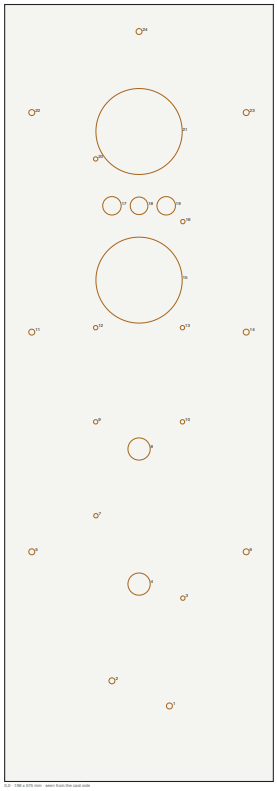


0,0 · 480 x 60 mm · seen from the outside (top)

#	x	y	Hole	For
1	474	11	4.5	leg holder
2	385	12	4.5	core post
3	95	12	4.5	core post
4	6	11	4.5	leg holder

Uprights

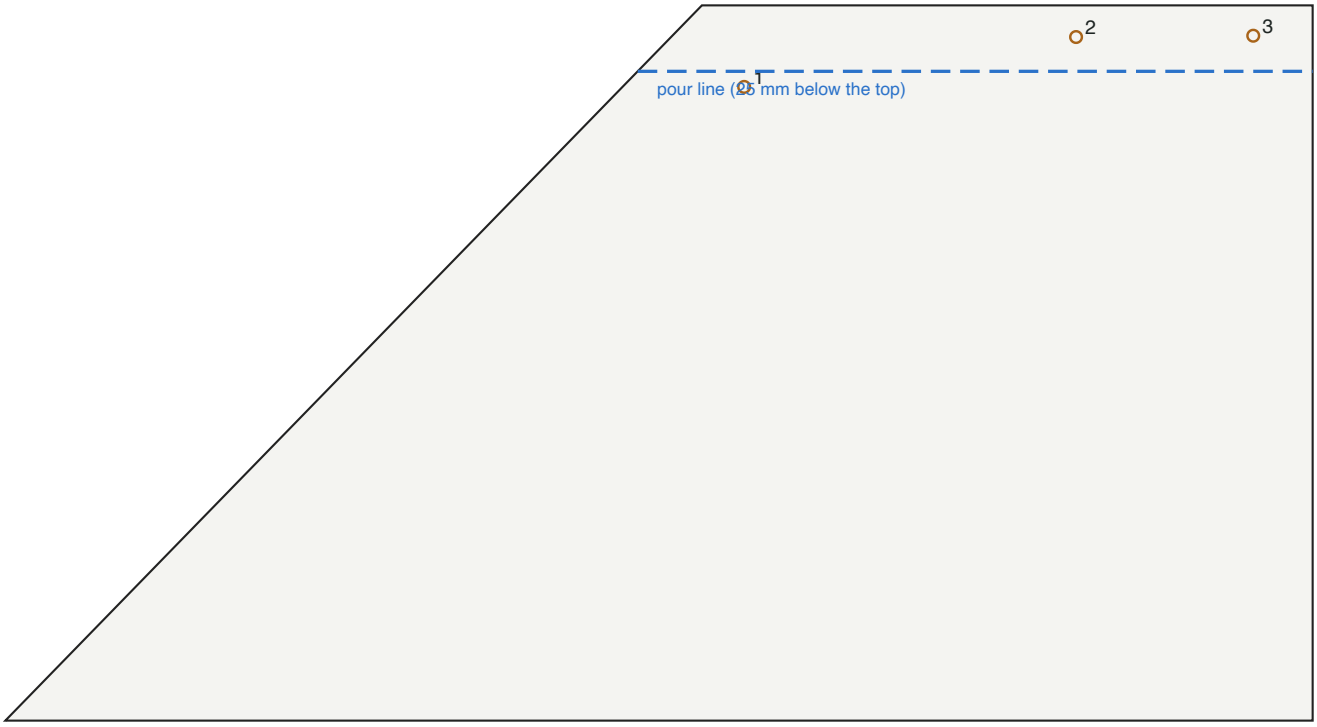
Upright floor (x2)



#	x	y	Hole	For
1	121	56	4.5	toe-filler
2	79	75	4.5	toe-filler
3	131	136	3.2	foot-gasket-groove
4	99	146	16.3	5/8 core rod

#	x	y	Hole	For
5	20	170	4.5	cleat
6	178	170	4.5	cleat
7	67	197	3.2	foot-gasket-groove
8	99	246	16.3	5/8 core rod
9	67	266	3.2	foot-gasket-groove
10	131	266	3.2	foot-gasket-groove
11	20	333	4.5	cleat
12	67	336	3.2	foot-gasket-groove
13	131	336	3.2	foot-gasket-groove
14	178	333	4.5	cleat
15	99	371	63.5	hole saw
16	131	414	3.2	foot-gasket-groove
17	79	426	13.5	1/2 rod through a sleeve
18	99	426	13.1	1/2-13 bolt, base dowel hole
19	119	426	13.5	1/2 rod through a sleeve
20	67	461	3.2	foot-gasket-groove
21	99	481	63.5	hole saw
22	20	495	4.5	cleat
23	178	495	4.5	cleat
24	99	555	4.5	cleat

Upright side plate, outer (x2)

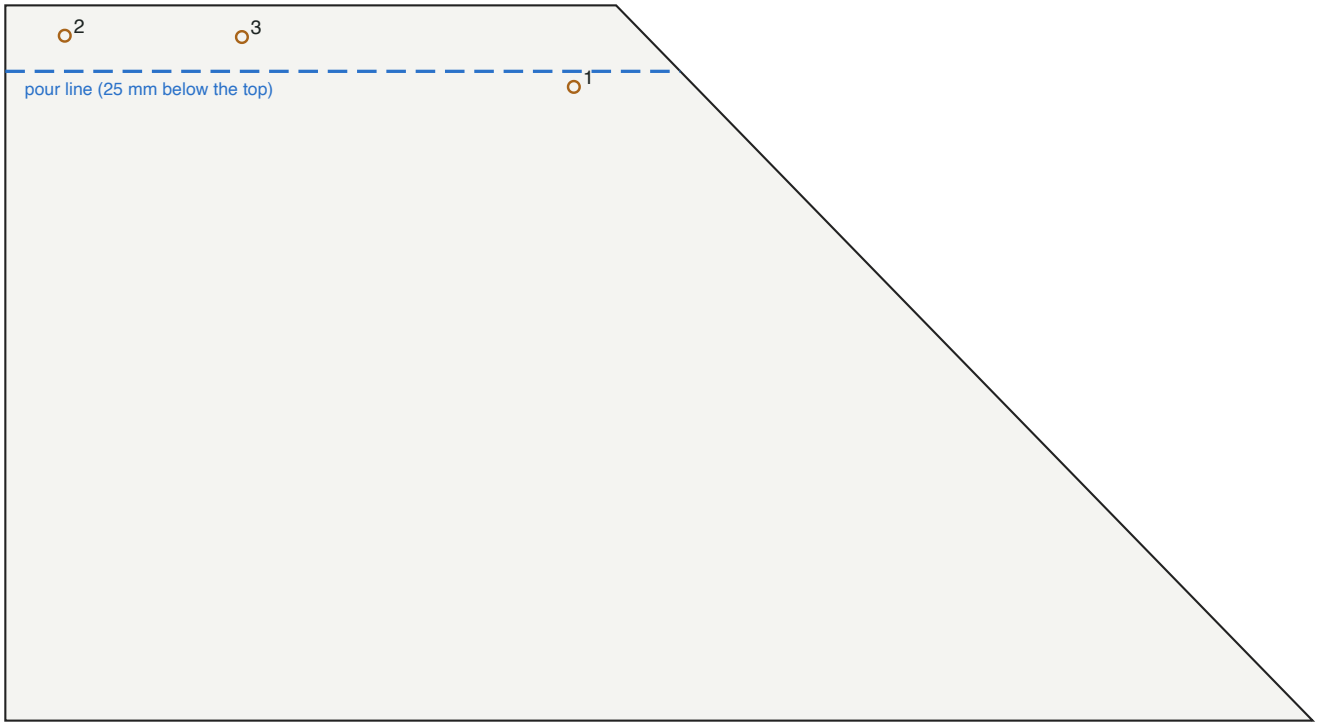


0,0 · 495 x 271 mm · seen from the cast side

#	x	y	Hole	For
1	280	240	4.5	corner bracket
2	406	259	4.5	saddle
3	473	260	4.5	corner bracket

Outline corners (x, y): (0, 0), (495, 0), (495, 271), (264, 271)

Upright side plate, inner (x2)

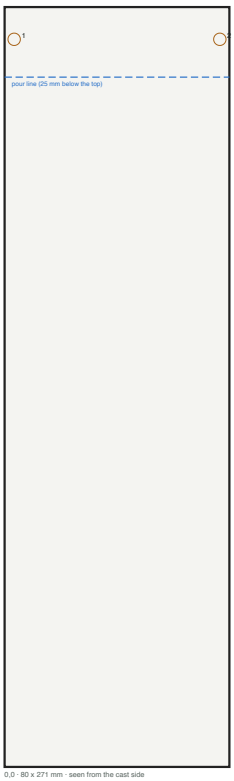


0,0 · 495 x 271 mm · seen from the cast side

#	x	y	Hole	For
1	215	240	4.5	corner bracket
2	22	260	4.5	corner bracket
3	90	259	4.5	saddle

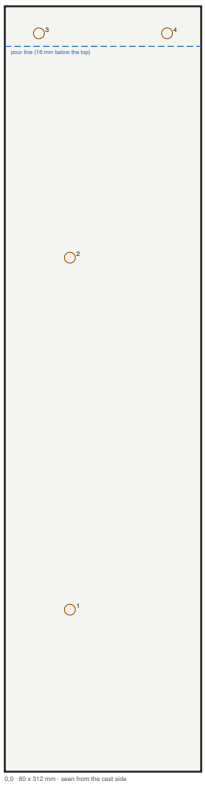
Outline corners (x, y): (495, 0), (0, 0), (0, 271), (231, 271)

Upright back board (x2)



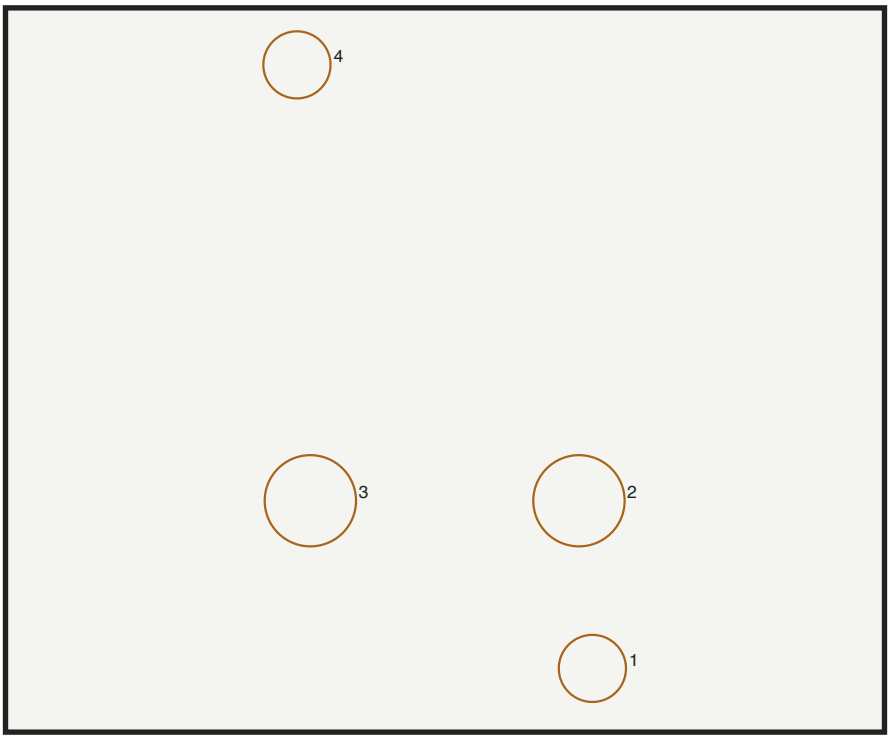
#	x	y	Hole	For
1	3	260	4.5	corner bracket
2	77	260	4.5	corner bracket

Upright front board (x2)



#	x	y	Hole	For
1	27	66	4.5	slope-plug-lower
2	27	209	4.5	slope-plug-upper
3	14	301	4.5	corner bracket
4	66	301	4.5	corner bracket

Upright bridge (x2)



0,0 · 131 x 108 mm · seen from the outside (top)

#	x	y	Hole	For
1	87	10	10	3/8-16 bolt into the top plate
2	85	34	13.5	1/2 rod through a sleeve
3	45	34	13.5	1/2 rod through a sleeve
4	43	100	10	3/8-16 bolt into the top plate

Upright cleat, outer (x2)



0,0 · 405 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	365	20	4.5	cleat
2	203	20	4.5	cleat
3	40	20	4.5	cleat

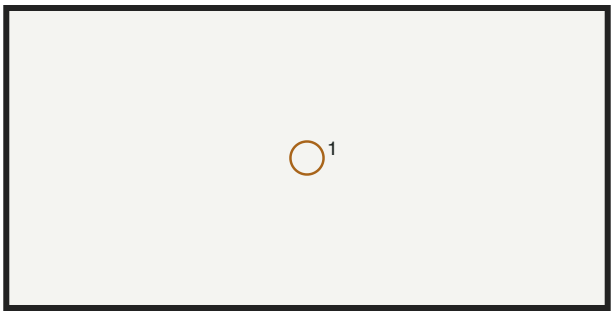
Upright cleat, inner (x2)



0,0 · 405 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	365	20	4.5	cleat
2	203	20	4.5	cleat
3	40	20	4.5	cleat

Upright cleat, back (x2)

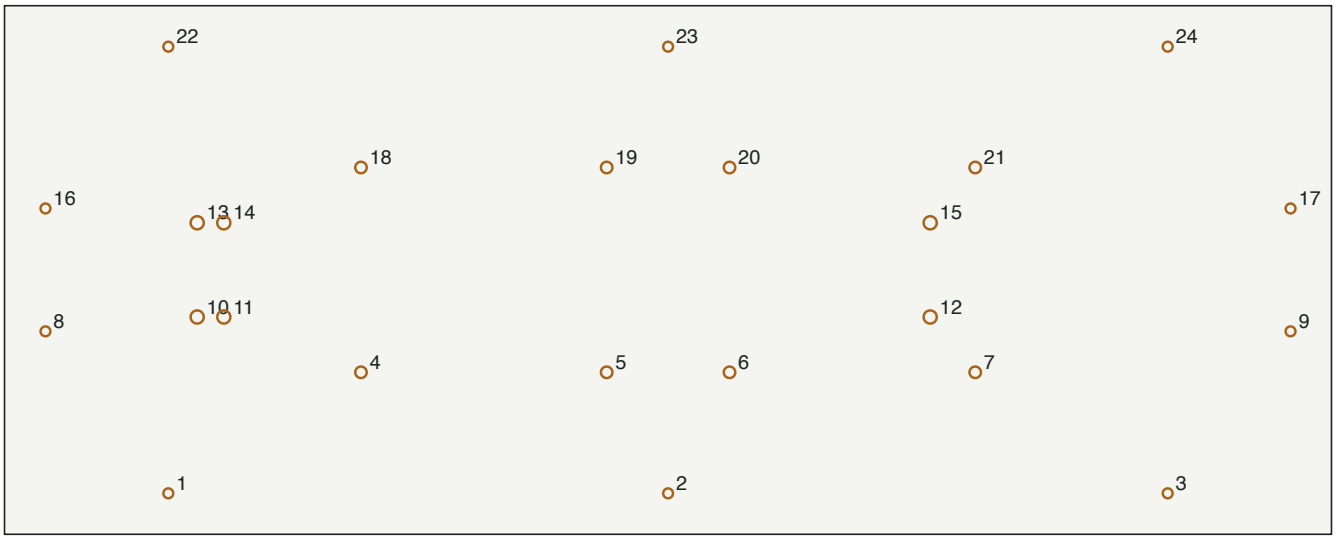


0,0 · 80 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	40	20	4.5	cleat

Beam

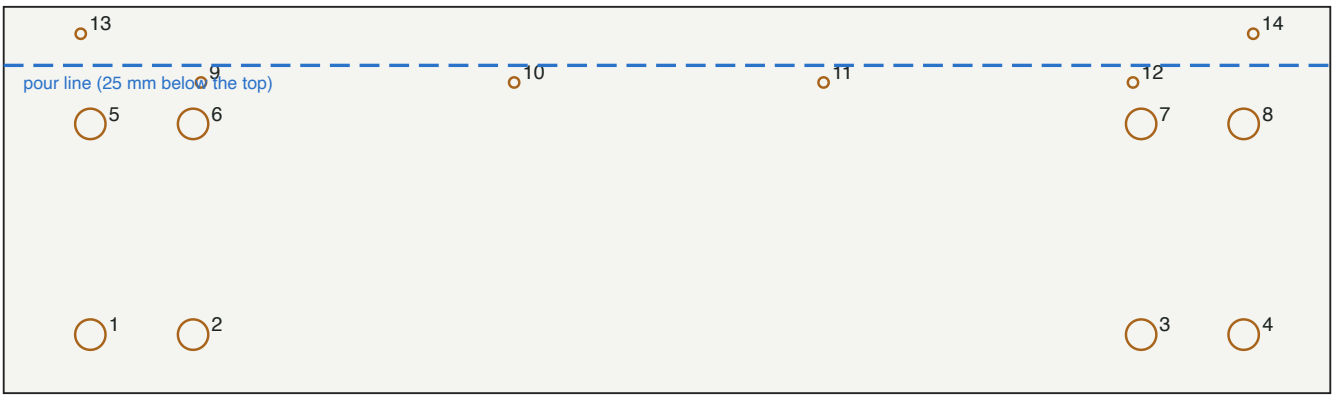
Beam floor



0,0 · 648 x 258 mm · seen from the cast side

#	x	y	Hole	For
1	80	20	4.5	cleat
2	324	20	4.5	cleat
3	568	20	4.5	cleat
4	174	79	5.5	M5 insert screw
5	294	79	5.5	M5 insert screw
6	354	79	5.5	M5 insert screw
7	474	79	5.5	M5 insert screw
8	20	99	4.5	cleat
9	628	99	4.5	cleat
10	94	106	6.5	M6 insert screw
11	107	106	6.5	M6 insert screw
12	452	106	6.5	M6 insert screw
13	94	152	6.5	M6 insert screw
14	107	152	6.5	M6 insert screw
15	452	152	6.5	M6 insert screw
16	20	159	4.5	cleat
17	628	159	4.5	cleat
18	174	179	5.5	M5 insert screw
19	294	179	5.5	M5 insert screw
20	354	179	5.5	M5 insert screw
21	474	179	5.5	M5 insert screw
22	80	238	4.5	cleat
23	324	238	4.5	cleat
24	568	238	4.5	cleat

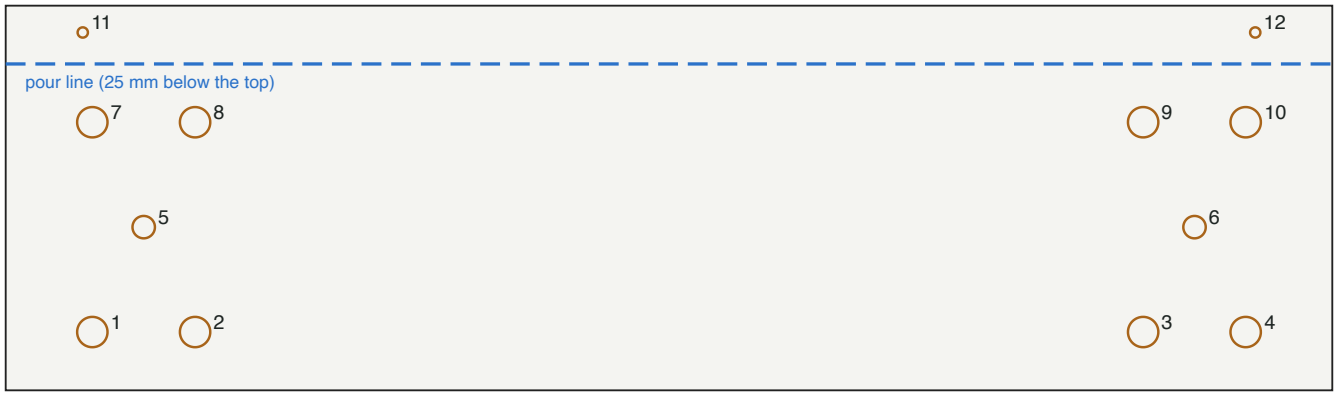
Beam top wall



0,0 · 568 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	37	25	13.1	1/2 core rod
2	81	25	13.1	1/2 core rod
3	487	25	13.1	1/2 core rod
4	531	25	13.1	1/2 core rod
5	37	115	13.1	1/2 core rod
6	81	115	13.1	1/2 core rod
7	487	115	13.1	1/2 core rod
8	531	115	13.1	1/2 core rod
9	84	133	4.5	back-edge
10	219	133	4.5	back-edge
11	351	133	4.5	back-edge
12	484	133	4.5	back-edge
13	33	154	4.5	corner bracket
14	535	154	4.5	corner bracket

Beam bottom wall

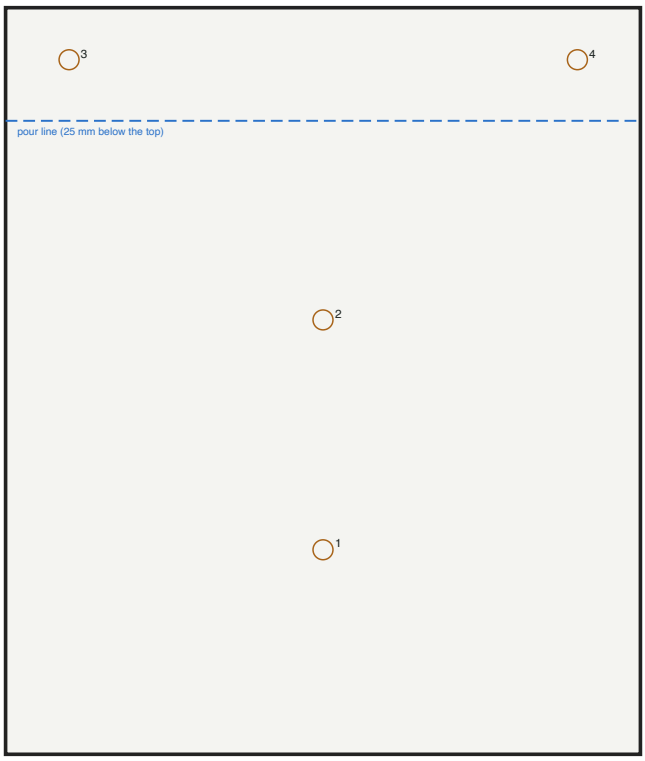


0,0 · 568 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	37	25	13.1	1/2 core rod
2	81	25	13.1	1/2 core rod
3	487	25	13.1	1/2 core rod
4	531	25	13.1	1/2 core rod
5	59	70	9.6	3/8 dowel
6	509	70	9.6	3/8 dowel
7	37	115	13.1	1/2 core rod
8	81	115	13.1	1/2 core rod

#	x	y	Hole	For
9	487	115	13.1	1/2 core rod
10	531	115	13.1	1/2 core rod
11	33	154	4.5	corner bracket
12	535	154	4.5	corner bracket

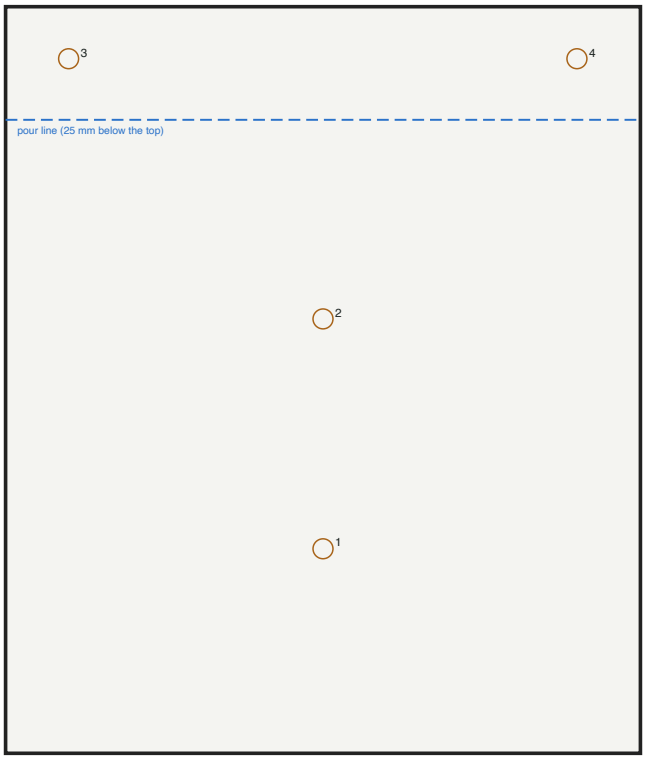
Beam end wall, left



0,0 · 140 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	70	45	4.5	tube locator
2	70	96	4.5	tube locator
3	14	154	4.5	corner bracket
4	126	154	4.5	corner bracket

Beam end wall, right



0,0 · 140 x 165 mm · seen from the cast side

#	x	y	Hole	For
1	70	45	4.5	tube locator
2	70	96	4.5	tube locator
3	14	154	4.5	corner bracket
4	126	154	4.5	corner bracket

Beam cleat, top



0,0 · 568 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	528	20	4.5	cleat
2	284	20	4.5	cleat
3	40	20	4.5	cleat

Beam cleat, bottom



0,0 · 568 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	528	20	4.5	cleat
2	284	20	4.5	cleat
3	40	20	4.5	cleat

Beam cleat, left



0,0 · 140 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	100	20	4.5	cleat
2	40	20	4.5	cleat

Beam cleat, right



0,0 · 140 x 40 mm · seen from the outside (top)

#	x	y	Hole	For
1	100	20	4.5	cleat
2	40	20	4.5	cleat